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PERSONALIZED EVENT DETECTION METHODS AND RELATED DEVICES AND SYSTEMS

Abstract

Techniques for detection of occurrence of a meal are provided. In some embodiments, the techniques may involve obtaining a plurality of glucose measurements indicative of a glucose level in a body of a patient during an analysis interval. The techniques may further involve detecting an occurrence of a meal during the analysis interval based on the plurality of glucose measurements obtained during the analysis interval. The techniques may further involve updating a user interface to include an indication of the meal detected during the analysis interval.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. patent application Ser. No. 18/749,277, filed Jun. 20, 2024 and titled “PERSONALIZED EVENT DETECTION,” which is a continuation of U.S. patent application Ser. No. 17/652,816, filed Feb. 28, 2022 and titled “PERSONALIZED EVENT DETECTION,” which is a continuation of U.S. patent application Ser. No. 16/579,651, filed Sep. 23, 2019 and titled “PERSONALIZED EVENT DETECTION METHODS AND RELATED DEVICES AND SYSTEMS,” which is a continuation of U.S. patent application Ser. No. 15/240,711, filed Aug. 18, 2016 and titled “PERSONALIZED EVENT DETECTION METHODS AND RELATED DEVICES AND SYSTEMS,” which claims the benefit of the following United States Provisional Patent Applications: U.S. Provisional Patent Application Ser. No. 62/208,479, filed Aug. 21, 2015 and titled “SENSOR CORE DATA SCIENCE”; U.S. Provisional Patent Application Ser. No. 62/266,820, filed Dec. 14, 2015 and titled “MEAL DETECTION MODEL AND IMPLEMENTATION FOR USE IN MOBILE APPLICATIONS AND PREDICTIVE ANALYTICS”; U.S. Provisional Patent Application Ser. No. 62/286,828, filed Jan. 25, 2016 and titled “GLYCEMIC INSIGHT ALGORITHMS AND IMPLEMENTATION FOR USE IN MOBILE APPLICATIONS AND PREDICTIVE ANALYTICS”; U.S. Provisional Patent Application Ser. No. 62/304,605, filed Mar. 7, 2016 and titled “GLYCEMIC INSIGHT ALGORITHMS AND IMPLEMENTATION FOR USE IN MOBILE APPLICATIONS AND PREDICTIVE ANALYTICS”; U.S. Provisional Patent Application Ser. No. 62/304,609, filed Mar. 7, 2016 and titled “INSIGHT GENERATION LOGIC BREAK DOWN (CURRENT LOGIC); U.S. Provisional Patent Application Ser. No. 62/304,615, filed Mar. 7, 2016 and titled “RETROSPECTIVE INSIGHT ALGORITHMS AND IMPLEMENTATION FOR DIABETES PATIENT ENGAGEMENT IN MOBILE APPLICATIONS”; U.S. Provisional Patent Application Ser. No. 62/304,618 filed Mar. 7, 2016 and titled “LIST OF ALL INSIGHTS”; and U.S. Provisional Patent Application Ser. No. 62/329,021, filed Apr. 28, 2016 and titled “GLUCOSE

ASSIST' SYSTEM." Each of these applications is incorporated by reference herein in its entirety and for all purposes.

TECHNICAL FIELD

[0002] Subject matter described herein relates generally to personalized event detection.

BACKGROUND

[0003] Infusion pump devices and systems are relatively well known in the medical arts, for use in delivering or dispensing an agent, such as insulin or another prescribed medication, to a patient. A typical infusion pump includes a pump drive system which typically includes a small motor and drive train components that convert rotational motor motion to a translational displacement of a plunger (or stopper) in a reservoir that delivers medication from the reservoir to the body of a user via a fluid path created between the reservoir and the body of a user. Use of infusion pump therapy has been increasing, especially for delivering insulin for diabetics.

[0004] Control schemes have been developed that allow insulin infusion pumps to monitor and regulate a user's blood glucose level in a substantially continuous and autonomous manner. However, regulating blood glucose level is still complicated by variations in the response time for the type of insulin being used along with variations in a user's individual insulin response and daily activities (e.g., exercise, carbohydrate consumption, bolus administration, and the like).

Additionally, manually-initiated deliveries of insulin prior to or contemporaneously with consuming a meal (e.g., a meal bolus or correction bolus) also influence the overall glucose regulation, along with various patient-specific ratios, factors, or other control parameters.

[0005] Continuous monitoring provides a greater understanding of the condition of a patient with diabetes. That said, there is also a burden imposed on patients, physicians and other healthcare providers to adapt to continuous monitoring and incorporate the amount of available data in a manner that allows for improved patient outcomes. While meals can be a major contributor to blood glucose variations, manual input is generally relied on to identify when meals occur and what the corresponding glucose response. However, many patients find manually indicating meals to be unnecessarily burdensome. In other instances, a patient may simply forget to provide a meal indication or be unable to provide a meal indication due to time or social constraints. Thus, for any given patient, a relatively large number of meals may occur without any indication that facilitates further analysis or monitoring. Accordingly, there is a need to identify meals and corresponding glucose responses without relying on manual interaction.

BRIEF SUMMARY

[0006] Disclose herein are techniques related to personalized event detection. The techniques may be practiced using a processor-implemented method; a system comprising one or more processors and one or more processor-readable storage media; and/or one or more non-transitory processor-readable storage media.

[0007] In some embodiments, the techniques may involve obtaining a plurality of glucose measurements indicative of a glucose level in a body of a patient during an analysis interval. The techniques may further involve detecting an occurrence of a meal during the analysis interval based on the plurality of glucose measurements obtained during the analysis interval. The techniques may further involve updating a user interface to include an indication of the meal detected during the analysis interval.

[0008] In some embodiments, the techniques may involve obtaining a plurality of glucose measurements indicative of a glucose level in a body of a patient during an analysis interval. The techniques may further involve detecting an occurrence of a meal during the analysis interval based on the plurality of glucose measurements obtained during the analysis interval. The techniques may further involve presenting a meal marker icon in a user interface that includes a graph presenting glucose measurements including the plurality of glucose measurements, wherein the meal marker icon is overlaid on the graph.

[0009] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A more complete understanding of the subject matter may be derived by referring to the detailed description and claims when considered in conjunction with the following figures, wherein like reference numbers refer to similar elements throughout the figures, which may be illustrated for simplicity and clarity and are not necessarily drawn to scale.

[0011] FIG. 1 depicts an exemplary embodiment of a patient management system;

[0012] FIG. 2 is a flow diagram of an exemplary patient modeling process suitable for use with the patient management system of FIG. 1 in one or more exemplary embodiments;

[0013] FIG. 3 is a graph depicting an exemplary relationship of sensor glucose measurement values with respect to time in accordance with one embodiment suitable for use with the patient modeling process of FIG. 2 in one or more exemplary embodiments;

[0014] FIG. 4 is a graph depicting an exemplary relationship of patient meal probability with respect to time in accordance with one embodiment suitable for use with the patient modeling process of FIG. 2 in one or more exemplary embodiments;

[0015] FIG. 5 is a table depicting an exemplary meal detection models generated for different patients in conjunction with the patient modeling process of FIG. 2 in one or more exemplary embodiments;

[0016] FIG. 6 is a flow diagram of an exemplary meal detection process suitable for use with a meal detection model generated by the patient modeling process of FIG. 2 in the patient management system of FIG. 1 in one or more exemplary embodiments;

[0017] FIG. 7 depicts an exemplary embodiment of a snapshot graphical user interface (GUI) display that may be presented on a display device associated with a computing device in one or more embodiments;

[0018] FIG. 8 depicts an embodiment of a computing device for a diabetes data management system in accordance with one or more embodiments;

[0019] FIG. 9 depicts an exemplary embodiment of an infusion system;

[0020] FIG. 10 depicts a plan view of an exemplary embodiment of a fluid infusion device suitable for use in the infusion system of FIG. 9;

[0021] FIG. 11 is an exploded perspective view of the fluid infusion device of FIG. 10;

[0022] FIG. 12 is a cross-sectional view of the fluid infusion device of FIGS. 10-11 as viewed along line 12-12 in FIG. 11 when assembled with a reservoir inserted in the infusion device;

[0023] FIG. 13 is a block diagram of an exemplary control system suitable for use in a fluid infusion device in one or more embodiments;

[0024] FIG. 14 is a block diagram of an exemplary pump control system suitable for use in the control system of FIG. 13 in one or more embodiments; and

[0025] FIG. 15 is a block diagram of a closed-loop control system that may be implemented or otherwise supported by the pump control system in the fluid infusion device of FIG. 13 in one or more exemplary embodiments.

DETAILED DESCRIPTION

[0026] The following detailed description is merely illustrative in nature and is not intended to limit the embodiments of the subject matter or the application and uses of such embodiments. As used herein, the word “exemplary” means “serving as an example, instance, or illustration.” Any

implementation described herein as exemplary is not necessarily to be construed as preferred or advantageous over other implementations. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description.

[0027] Exemplary embodiments of the subject matter described herein are implemented in conjunction with medical devices, such as portable electronic medical devices. Although many different applications are possible, the following description focuses on embodiments that incorporate a fluid infusion device (or infusion pump) as part of an infusion system deployment. For the sake of brevity, conventional techniques related to infusion system operation, insulin pump and/or infusion set operation, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail here. Examples of infusion pumps may be of the type described in, but not limited to, U.S. Pat. Nos. 4,562,751; 4,685,903; 5,080,653; 5,505,709; 5,097,122; 6,485,465; 6,554,798; 6,558,320; 6,558,351; 6,641,533; 6,659,980; 6,752,787; 6,817,990; 6,932,584; and 7,621,893; each of which are herein incorporated by reference. That said, the subject matter described herein can be utilized more generally in the context of overall diabetes management or other physiological conditions independent of or without the use of an infusion device or other medical device (e.g., when oral medication is utilized), and the subject matter described herein is not limited to any particular type of medication.

[0028] Generally, a fluid infusion device includes a motor or other actuation arrangement that is operable to linearly displace a plunger (or stopper) of a reservoir provided within the fluid infusion device to deliver a dosage of fluid, such as insulin, to the body of a user. Dosage commands that govern operation of the motor may be generated in an automated manner in accordance with the delivery control scheme associated with a particular operating mode, and the dosage commands may be generated in a manner that is influenced by a current (or most recent) measurement of a physiological condition in the body of the user. For example, in a closed-loop operating mode, dosage commands may be generated based on a difference between a current (or most recent) measurement of the interstitial fluid glucose level in the body of the user and a target (or reference) glucose value. In this regard, the rate of infusion may vary as the difference between a current measurement value and the target measurement value fluctuates. For purposes of explanation, the subject matter is described herein in the context of the infused fluid being insulin for regulating a glucose level of a user (or patient); however, it should be appreciated that many other fluids may be administered through infusion, and the subject matter described herein is not necessarily limited to use with insulin.

[0029] Exemplary embodiments described herein generally relate to systems for analyzing and presenting information pertaining to operation of an infusion device delivering fluid to a body of a user, and in particular, information pertaining to events occurring during operation. For example, in one or more embodiments, a snapshot graphical user interface (GUI) display may be presented that includes graphical indicia of event pattern(s) identified during preceding operation of the infusion device. The event pattern information may be utilized by the patient, the patient's doctor or other health care provider, or another individual to assess the efficacy of the regulation achieved by the infusion device, identify potential causes of an event pattern, and identify potential actions that may improve the quality of control achieved by the infusion device.

[0030] In exemplary embodiments described herein, events occurring during operation of an infusion device are capable of being detected autonomously and without reliance on any manual interaction or inputs using a personalized, patient-specific event detection model. The event detection model relies on input variables that have been identified as predictive or correlative to an occurrence of the event for that patient, which may include patient-specific event probabilities determined based on historical event data associated with the patient and one or more measurement statistics determined based on measurement data for a physiological condition of the patient. In this regard, for each patient, the particular subset of measurement statistics that are predictive or

correlative to the event for that patient may be different from those of other patients, and similarly, whether the historical event probability is predictive or correlative to the event may also vary on a patient-by-patient basis. Additionally, the patient-specific event detection model also utilizes correlation coefficient values to weight the relative predictiveness or correlative strength of each model input variable. Thus, the model determines a metric indicative of the likelihood of an occurrence or an absence of the event based on the patient-specific predictive subset of measurement statistics and corresponding patient-specific correlation coefficient values. In some embodiments, events may be identified substantially in real-time.

[0031] In exemplary embodiments, the events to be detected are meals occurring during monitoring of a patient's glucose levels and corresponding operation of an insulin infusion device. In this regard, a patient-specific meal detection model is employed to calculate or otherwise determine a metric indicative of whether or not a meal corresponding to a current analysis interval has occurred based on a subset of glucose measurement statistics that have been identified as predictive of or correlative to consumption of a meal by the patient based on the patient's historical meal and bolus information. For some patients, the meal detection model may also incorporate a meal probability metric for the current analysis interval that reflects the likelihood of the patient consuming a meal contemporaneously or concurrently to the current analysis interval based on the patient's historical meal distribution. Values for the predictive subset of glucose measurement statistics are calculated based on glucose measurement data corresponding to the current analysis interval and a meal probability metric for the current analysis interval is also determined, and the respective correlation coefficients from the meal detection model are applied to the current values for the predictive subset of glucose measurement statistics and the current meal probability to obtain a meal consumption metric value. The meal consumption metric value is then utilized to determine whether a meal has been consumed during the current analysis interval.

[0032] When consumption of a meal is detected, the glucose measurement data associated with the current analysis interval may be flagged, marked, or otherwise stored in association with an identifier that indicates a meal associated with the analysis interval corresponding to that subset of the glucose measurement data. Accordingly, any notifications, alerts, displays, or other actions that would otherwise be undertaken in response to a manually-indicated or announced meal may be similarly performed in response to or in a manner that is influenced by a meal detected by the meal detection model. Thus, meals may be autonomously identified and automatically reflected by subsequent operation of the infusion device or displays pertaining thereto without requiring any manual interaction. Not only is the patient alleviated of the burdens associated with manually announcing meals, but inadvertently omitted meal indications may also be accounted for. As a result, the number of meals for which data may be captured and analyzed may increase, which, in turn, may increase the accuracy or reliability of meal response-related analysis, and thereby improve patient outcomes.

[0033] FIG. 1 depicts an exemplary embodiment of a patient management system **100**. The patient management system **100** includes an infusion device **102** that is communicatively coupled to a sensing arrangement **104** to obtain measurement data indicative of a physiological condition in the body of a patient, such as sensor glucose measurement values, as described in greater detail below in the context of FIGS. 9-12. In exemplary embodiments, the infusion device **102** operates autonomously to regulate the patient's glucose level based on the sensor glucose measurement values received from the sensing arrangement **104** as described in greater detail below in the context of FIGS. 13-15.

[0034] In the illustrated embodiment, the infusion device **102** periodically uploads or otherwise transmits the measurement data (e.g., sensor glucose measurement values and timestamps associated therewith) to a remote device **106** via a communications network **114**, such as a wired and/or wireless computer network, a cellular network, a mobile broadband network, a radio network, or the like. That said, in other embodiments, the sensing arrangement **104** may be

communicatively coupled to the communications network **114** to periodically upload or otherwise transmit measurement data to the remote device **106** via the communications network **114** independent of the infusion device **102**. Additionally, in some embodiments, the infusion device **102** also uploads delivery data and/or other information indicative of the amount of fluid delivered by the infusion device and the timing of fluid delivery, which may include, for example, information pertaining to the amount and timing of manually-initiated boluses and associated meal announcements. Some examples of an infusion device uploading measurement and delivery data to a remote device are described in United States Patent Application Publication Nos. 2015/0057807 and 2015/0057634, which are incorporated by reference herein in their entirety.

[0035] The remote device **106** is coupled to a database **108** configured to store or otherwise maintain the historical measurement and delivery data received from the infusion device **102** in association with a patient associated with the infusion device **102** (e.g., using unique patient identification information). Additionally, the database **108** may store or otherwise maintain, in association with a particular patient, a personalized and patient-specific event detection model. In this regard, the event detection model defines which input variables or parameters are to be factored in when making a determination of whether or not an event has occurred along with relative weightings assigned to those inputs corresponding to how predictive or correlative the value of a respective model input value is to the occurrence of the event to be detected, as described in greater detail below in the context of FIGS. 2-6. The remote device **106** generally represents an electronic device configured to analyze or otherwise monitor the measurement and delivery data obtained for the patient associated with the infusion device **102**, generate patient-specific event detection models based on a respective patient's historical measurement and delivery data, and generate or otherwise facilitate a GUI display that is influenced by or otherwise reflects the detected events. The GUI display may be presented on the remote device **106** or another electronic device **110**, alternatively referred to herein as a client device. In practice, the remote device **106** may reside at a location that is physically distinct and/or separate from the infusion device **102**, such as, for example, at a facility that is owned and/or operated by or otherwise affiliated with a manufacturer of the infusion device **102**. For purposes of explanation, but without limitation, the remote device **106** may alternatively be referred to herein as a server.

[0036] The remote device **106** generally represents a computing system or another combination of processing logic, circuitry, hardware, and/or other components configured to support the processes, tasks, operations, and/or functions described herein. In this regard, the server **106** includes a processing system **116**, which may be implemented using any suitable processing system and/or device, such as, for example, one or more processors, central processing units (CPUs), controllers, microprocessors, microcontrollers, processing cores and/or other hardware computing resources configured to support the operation of the processing system **116** described herein. The processing system **116** may include or otherwise access a data storage element **118** (or memory) capable of storing programming instructions for execution by the processing system **116**, that, when read and executed, cause processing system **116** to perform or otherwise support the processes, tasks, operations, and/or functions described herein. For example, in one embodiment, the instructions cause the processing system **116** to create, generate, or otherwise facilitate an application platform that supports instances of an application using data that is stored or otherwise maintained by the database **108**. Depending on the embodiment, the memory **118** may be realized as a random access memory (RAM), read only memory (ROM), flash memory, magnetic or optical mass storage, or any other suitable non-transitory short or long term data storage or other computer-readable media, and/or any suitable combination thereof.

[0037] The client device **110** generally represents an electronic device coupled to the network **114** that may be utilized by a user to access and view data stored in the database **108** via the server **106**. In practice, the client device **110** can be realized as any sort of personal computer, mobile telephone, tablet or other network-enabled electronic device that includes a display device, such as

a monitor, screen, or another conventional electronic display, capable of graphically presenting data and/or information provided by the server **106** along with a user input device, such as a keyboard, a mouse, a touchscreen, or the like, capable of receiving input data and/or other information from the user of the client device **110**. A user, such as the patient's doctor or another healthcare provider, manipulates the client device **110** to execute a client application **112** that contacts the server **106** via the network **114** using a networking protocol, such as the hypertext transport protocol (HTTP) or the like.

[0038] In exemplary embodiments described herein, a user of the client device **110** manipulates a user input device associated with the client device **110** to input or otherwise provide indication of the patient associated with the infusion device **102** along with a period of time for which the user would like to review, analyze, or otherwise assess measurement data associated with the patient. In response, the server **106** accesses the database **108** to retrieve or otherwise obtain historical measurement data associated with the identified patient for the identified time period and generates a GUI display that is presented on the display device associated with the client device **110** via the client application **112** executing thereon.

[0039] It should be appreciated that FIG. **1** depicts a simplified representation of a patient management system **100** for purposes of explanation and is not intended to limit the subject matter described herein in any way. For example, in various embodiments, a GUI display may be presented on any device within the patient management system **100** (e.g., the server **106**, the infusion device **102**, the sensing arrangement **104**, or the like) and not necessarily on the client device **110**. Moreover, in some embodiments, the infusion device **102** may be configured to store or otherwise maintain historical measurement and delivery data onboard the infusion device **102** and generate GUI displays on a display device associated with the infusion device **102**, in which case one or more of the server **106**, the database **108**, and the client device **110** may not be present. Additionally, the subject matter described herein is not limited to meal detection or event detection for purposes of generating GUI displays, and in various embodiments, meal or event detection may be utilized to dynamically adjust operation or otherwise influence the control scheme of the infusion device **102** (e.g., by adjusting one or more thresholds or parameters influencing delivery, notifications, alerting, or the like), generate alerts or other user notifications, or perform other actions in response to the detected meal or event.

[0040] FIG. **2** depicts an exemplary patient modeling process **200** suitable for implementation by a patient management system to develop a patient-specific event detection model. The various tasks performed in connection with the patient modeling process **200** may be performed by hardware, firmware, software executed by processing circuitry, or any combination thereof. For illustrative purposes, the following description refers to elements mentioned above in connection with FIG. **1**. In practice, portions of the patient modeling process **200** may be performed by different elements of the patient management system **100**, such as, for example, the infusion device **102**, the sensing arrangement **104**, the server **106**, the database **108**, the client device **110**, the client application **112**, and/or the processing system **116**. It should be appreciated that the patient modeling process **200** may include any number of additional or alternative tasks, the tasks need not be performed in the illustrated order and/or the tasks may be performed concurrently, and/or the patient modeling process **200** may be incorporated into a more comprehensive procedure or process having additional functionality not described in detail herein. Moreover, one or more of the tasks shown and described in the context of FIG. **2** could be omitted from a practical embodiment of the patient modeling process **200** as long as the intended overall functionality remains intact.

[0041] The patient modeling process **200** begins by obtaining historical measurement data for the patient of interest and obtaining historical bolus data for the patient over the period corresponding to the historical measurement data (tasks **202**, **204**). In this regard, the patient modeling process **200** may be initiated or performed after the server **106** has acquired and stored sufficient sensor glucose measurement values and bolus information in the database **108**. For example, the infusion

device **102** may periodically upload, to the server **106** via the network **114**, reference blood glucose measurement values obtained from the body of the patient (e.g., using a blood glucose meter or fingerstick device) along with bolus information including the timings and amounts of insulin delivered, including indications of whether a particular bolus is a meal bolus or otherwise associated with a meal. The bolus information may also include the amount of carbohydrates consumed, the type of meal, or the like. In this regard, in the absence of an explicit meal indication or announcement from the patient, the server **106** may automatically classify a bolus delivered as a meal bolus when a carbohydrate entry occurred within a threshold amount of time of the bolus being delivered (e.g., within 5 minutes). Additionally, the infusion device **102** (or alternatively, the sensing arrangement **104**) may periodically upload, to the server **106**, sensor glucose measurement values obtained from the body of the patient by the sensing arrangement **104**.

[0042] In one or more embodiments, the server **106** stores or otherwise maintains, in the database **108**, one or more files or entries associated with the patient that maintains an association between the patient's historical sensor glucose measurement data, the patient's historical bolus and meal data, the patient's historical reference blood glucose measurements, and the like. Once a sufficient amount of data has been obtained by the server **106** and/or the database **108**, the patient modeling process **200** continues with determining a patient-specific meal detection model based on the patient's historical sensor glucose measurement and bolus data. In one embodiment, the patient modeling process **200** requires that data for at least a minimum threshold number of days (or hours) has been uploaded to continue. In other embodiments, additional thresholds may be utilized to determine when modeling can occur, such as, for example, a minimum average number of meal boluses per day of data, a minimum average number of reference blood glucose values per day of data, or the like.

[0043] In some embodiments, the server **106** also performs or otherwise implements one or more data cleaning or data preparation processes on the data to reduce noise or transients within the data, such as, for example, low-pass filtering the sensor glucose measurement values or otherwise filtering, excluding, or mitigating the impact of portions of data accompanied by variables that could cause variations that would reduce accuracy or reliability of the model. For example, the server **106** may identify and reduce or otherwise eliminate the impact of data associated with periods of time involving anomalous measurement data, such as, for example, artifacts, dropouts, or the like. In some embodiments, the measurement data associated with or potentially influenced by artifacts, dropouts or other anomalous sensor behavior may be detected by either the sensing arrangement **104** or the infusion device **102** instead of the server **106**. Examples of artifact or dropout detection are described in U.S. Patent Application Pub. No. 2015/0328402 and incorporated by reference herein. In other embodiments, the data cleaning or filtering may also account for suspension of delivery by the infusion device **102** or meals accompanied by stacked boluses, anomalous or unusual carbohydrate amounts, anomalous or unusual bolus dosages, particular bolus types, anomalous or unusual carbohydrate ratios, anomalous or unusual sensitivity factors, anomalous or unusual glucose targets, anomalous or unusual basal rates, anomalous or unusual active insulin amounts, or the like.

[0044] Once sufficient historical measurement and bolus data is obtained, the patient modeling process **200** continues by classifying or categorizing segments of time within the period of time encompassed by the historical data into meal or non-meal time segments (task **206**). In an exemplary embodiment, a meal segment is defined as a one hour segment of historical sensor glucose measurement values that starts around the time of a meal bolus (at, before or after) and precedes the highest or maximum sensor glucose measurement value within the 90 minutes following the meal bolus, where that meal bolus is also associated with a carbohydrate amount greater than a minimum threshold (e.g., 0 grams), is delivered within a maximum amount of time difference (e.g., 5 minutes) before or after the carbohydrate entry, and is not preceded by another meal bolus within 3 hours of the reference meal bolus associated with the meal segment. In this

regard, the threshold criteria are intended to diminish the effects of active insulin from other boluses and ensure the meal segments reliably reflect a characteristic meal response of the patient. In exemplary embodiments, a non-meal segment is defined as a one hour segment of historical sensor glucose measurement values that precedes or follows a meal segment. Historical sensor glucose measurement values that are not classified into a meal or non-meal segment are not utilized for purposes of generating the model due to those segments of data potentially exhibiting responses to residual active insulin from preceding meal boluses, as well as maintaining segments of consistent duration. In one or more embodiments where an absolute timestamp is available for when a patient begins eating a meal (e.g., by a user indication via the client application **112**), a meal segment may be defined as the one hour segment of historical sensor glucose measurement values that starts at that time.

[0045] FIG. **3** depicts an exemplary graph **300** of sensor glucose measurement values with respect to time illustrating one example of how sensor glucose measurement data could be classified into different meal or non-meal segments based on meal bolus indications and the sensor glucose measurement values. In the illustrated example, segments **302**, **304**, **306** of sensor glucose measurement values around meal bolus indications **301**, **303**, **305** are classified as meal segments, with segments **310**, **312**, **314**, **316** and **318** preceding the respective meal segments **302**, **304**, **306** being classified as non-meal segments. Here, it is noted that intervening segments **320**, **322**, **324**, **326** between meal segments **302**, **304** and non-meal segments **312**, **314**, **316** do not satisfy the criteria for being classified as either a meal or non-meal segment, and therefore, are excluded from further use in developing a meal detection model.

[0046] Referring again to FIG. **2**, after classifying the historical sensor glucose measurement data for the patient into meal and non-meal segments, the patient modeling process **200** continues by calculating or otherwise determining sensor glucose measurement statistics associated with the respective segments based on the sensor glucose measurement values corresponding to that respective segment (task **208**). In this regard, the sensor glucose measurement statistics characterize the patient's glucose level within or during the respective segment. For example, in one embodiment, for each one hour meal or non-meal segment, the server **106** calculates a mean or average sensor glucose measurement value over that one hour period, a standard deviation of the sensor glucose measurement values during that one hour period, a mean or average rate of change of the sensor glucose measurement values on a sample-to-sample basis during that one hour period, mean or average rate of changes of the sensor glucose measurement values on a sample-to-sample basis for discrete time periods within that one hour period (e.g., within the first 30 minutes and the second 30 minutes), standard deviations associated with those rate of change values, an absolute amplitude of the sensor glucose measurement values during that segment (e.g., the difference between the maximum and minimum values within that segment), and an amplitude difference between the first and last glucose measurement value of that segment (e.g., by subtracting the first glucose measurement value from the last glucose measurement value).

[0047] The patient modeling process **200** also calculates or otherwise determines a meal probability associated with the respective segments based on the patient's bolus history (task **210**). In this regard, the server **106** analyzes the meal bolus indications associated with the patient to determine a meal probability distribution for the patient that represents the relative probability of the patient consuming a meal at a particular time of day. In one embodiment, the server **106** divides the period of a day into a plurality of intervals, with a respective meal probability associated with each interval being determined based on the number of meal bolus indications having an associated timestamp within that interval relative. For example, FIG. **4** depicts an exemplary meal distribution **400** with respect to the time of day divided into 30 minute intervals, resulting in 48 different intervals over a 24-hour period. The corresponding meal probability for each interval may be determined by dividing the number of meal boluses classified or assigned to that respective interval by the total number of meal boluses. Thereafter, each meal or non-meal segment may be assigned a

corresponding meal probability value based on the corresponding intervals overlapped or concurrent to a respective segment.

[0048] In one embodiment, meal bolus instances for a patient are grouped into 10 minute intervals, resulting in 144 different intervals over a 24-hour period. For example, for a meal segment **302** spanning from 11:30 AM to 12:30 PM, the server **106** may determine a meal probability associated with that meal segment **302** as a sum of the number of instances of meal boluses with the six ten minute intervals encompassed by the meal segment (e.g., the 11:30 AM-11:40 AM interval, the 11:40 AM-11:50 AM interval, the 11:50 AM-12:00 PM interval, and so on) and then dividing the resultant sum by the total number of instances of meal boluses in the patient's history. It should be noted that 10 minute intervals are depicted and described for purposes of explanation, but in practice, the meal probability intervals are not limited to any particular length or duration, and may be shorter or longer than 10 minutes to achieve a desired level of granularity.

[0049] Still referring to FIG. 2, after determining the sensor glucose measurement statistics and meal probability values associated with each respective meal segment or non-meal segment, the patient modeling process **200** continues by identifying a subset of the sensor glucose measurement statistics and meal probability values that are predictive or correlative to the occurrence of a meal for that individual patient and generating a patient-specific meal detection model for that patient using that predictive subset of variables (tasks **212**, **214**). In this regard, in exemplary embodiments, the server **106** utilizes machine learning to determine which of the sensor glucose measurement statistics and the meal probability value are most strongly correlated to or predictive of each segment being of the classified type, and then determines a corresponding equation for calculating a metric indicative of whether or not a segment is a meal segment based on that subset of variables. Thus, the model is capable of characterizing or mapping the patient's response evidenced by the sensor glucose measurement values to that of a meal, and vice versa. Since each patient's response may vary from the rest of the population, the subset of sensor glucose measurement statistics that are predictive of or correlative to a meal for that patient may vary from other users. Additionally, the relative weightings applied to the respective statistics of that predictive subset may also vary from other patient's who may have common predictive subsets, based on differing correlations between a particular glucose measurement statistic and the meal response of that particular patient.

[0050] In one embodiment, stepwise logistic regression is utilized by the server **106** to determine what sensor glucose measurement statistics among the plurality of sensor glucose measurement statistics determined at task **208** are predictive of a meal for the patient of interest, as well as identifying whether the meal probability values determined at task **210** for that patient are predictive. In this regard, the meal probability value may not be predictive or correlative for a patient that grazes or consumes meals throughout the day, while the meal probability value may be highly predictive or correlative for a patient that grazes or consumes meals at relatively defined times of day. In some embodiments, the relative strength of correlation for each of those predictive variables may be utilized to determining a weighting or scaling for the values of those variables to arrive at an output from the model (e.g., by applying those weightings to those variable values) that indicates the occurrence or non-occurrence of a meal during an analysis interval (e.g., a one hour segment of sensor glucose measurement values). In other embodiments, multiple predictive variables may be multiplied or otherwise factored in with one another when values for those variables, in combination, are predictive or correlative to a meal.

[0051] In one or more exemplary embodiments, only a subset of the meal and non-meal segments for the patient are used to develop the meal detection model, with the remaining the meal and non-meal segments being utilized by the patient modeling process **200** to test or otherwise validate the developed model (tasks **216**, **218**). For example, for segments classified into the testing group of segments, the server **106** applies the developed meal detection model to the predictive variable values associated with the respective test segment, and then identifies or otherwise determines

whether the model results in that test segment being classified as a meal segment or non-meal segment. Thereafter, the server **106** compares the model classification of each test segment to the previous classification (e.g., task **206**) and calculates or otherwise determines one or more metrics indicative of the performance of the model with respect to the test segments. For example, the server **106** may determine an accuracy metric associated with the model based on the number of times the model correctly classified a test segment as a meal or non-meal segment. The server **106** may also calculate a sensitivity metric associated with the model as the ratio of the correctly detected number of meals by the model relative to the total number of meal segments in the test group. The server **106** may also calculate a specificity metric associated with the model as the ratio of the correctly detected number of non-meal segments by the model relative to the total number of non-meal segments in the test group.

[0052] When the performance metrics associated with the developed model are greater than or otherwise satisfy applicable validation criteria, the patient modeling process **200** stores or otherwise maintains the model in association with the patient for use in subsequently detecting meals that are not manually indicated or announced by the patient (task **220**). For example, identification of the predictive variables for the patient along with the relative weightings or manner in which those predictive variables should be combined to obtain a metric indicative of a meal may be stored or otherwise maintained in the database **108** in association with a patient identifier assigned to or otherwise associated with the patient and/or the infusion device **102**. Conversely, when the performance metrics associated with the developed model do not satisfy applicable validation criteria, the patient modeling process **200** discards the developed model and assigns or otherwise associates the patient with a broader population model (task **222**). In this regard, the population model may be developed by performing various aspects of the patient modeling process **200** across meal and non-meal segments associated with a plurality of different patients. For example, in one embodiment, a patient may be assigned or otherwise associated with a particular group of patients having one or more characteristics in common, with a meal detection model for that patient group being determined based on meal and non-meal segments for the different patients of the group. In one or more embodiments, the patient modeling process **200** assigns or otherwise associates the patient with a patient group meal detection model upon initialization of the patient within the patient management system **100** prior to accumulating sufficient historical data for developing a patient-specific model.

[0053] FIG. 5 depicts a table **500** illustrating different meal detection models generated for different patients. A first patient (Patient 1) has 364 days worth of historical measurement and bolus data stored in the database **108** that includes 72,660 instances of sensor glucose measurement values that can be classified into 1028 meal segments and 4936 non-meal segments, which are further divided into a training group of segments (consisting of 813 meal segments and 3833 non-meal segments) and a test group of segments (consisting of 215 meal segments and 1103 non-meal segments). The server **106** analyzes the test group of segments and determines a patient-specific meal detection model **502** for Patient 1 that generates a metric indicative of a meal occurrence based on the predictive variables (or model inputs) of: mean sensor glucose rate of change over the preceding one hour of sensor glucose measurement values, mean sensor glucose rate of change over the first half hour of the preceding one hour of sensor glucose measurement values, the meal probability value associated with the preceding one hour interval, the product of the timestamp associated with the meal and the amplitude difference of the sensor glucose measurement value over the preceding one hour interval, and the product of the mean and standard deviation of the sensor glucose rate of change over the second half hour of the preceding one hour of sensor glucose measurement values. After determining the model **502**, the server **106** applies the model **502** to the testing group of segments and then determines performance metrics associated with the meal/non-meal outputs model **502**. When the performance metrics satisfy validation criteria, the server **106** stores or otherwise maintains information defining the model **502** in the database **108** in association

with Patient 1.

[0054] A second patient (Patient 2) has 5 days worth of historical measurement and bolus data stored in the database **108** that includes 967 instances of sensor glucose measurement values that can be classified into 10 meal segments and 70 non-meal segments, which are further divided into a training group of segments (consisting of 5 meal segments and 55 non-meal segments) and a test group of segments (consisting of 5 meal segments and 15 non-meal segments). The server **106** analyzes the test group of segments and determines a patient-specific meal detection model **504** for Patient 2 that generates a metric indicative of a meal occurrence based on the predictive variables (or model inputs) of: standard deviation of the sensor glucose rate of change over the first half hour of the preceding one hour of sensor glucose measurement values and the meal probability value associated with the one hour interval. After determining the model **504**, the server **106** applies the model **504** to the testing group of segments and then determines performance metrics associated with the meal/non-meal outputs of the model **504**. In this regard, if the performance metrics satisfy validation criteria, the server **106** discards the model **504** or otherwise fails to store the model **504** in association with Patient 2. For example, if the validation criteria are an accuracy greater than 95%, a sensitivity greater than 80%, and a specificity greater than 80%, the Patient 1 model **502** may be validated and stored in association with Patient 1 while the Patient 2 model **504** fails the validation and is discarded or otherwise ignored in favor of a more generic population model for Patient 2.

[0055] FIG. **6** depicts an exemplary meal detection process **600** suitable for implementation by a patient management system to detect or otherwise identify unannounced meals using a patient-specific meal detection model. The various tasks performed in connection with the meal detection process **600** may be performed by hardware, firmware, software executed by processing circuitry, or any combination thereof. For illustrative purposes, the following description refers to elements mentioned above in connection with FIG. **1**. In practice, portions of the meal detection process **600** may be performed by different elements of the patient management system **100**, such as, for example, the infusion device **102**, the sensing arrangement **104**, the server **106**, the database **108**, the client device **110**, the client application **112**, and/or the processing system **116**. It should be appreciated that the meal detection process **600** may include any number of additional or alternative tasks, the tasks need not be performed in the illustrated order and/or the tasks may be performed concurrently, and/or the meal detection process **600** may be incorporated into a more comprehensive procedure or process having additional functionality not described in detail herein. Moreover, one or more of the tasks shown and described in the context of FIG. **6** could be omitted from a practical embodiment of the meal detection process **600** as long as the intended overall functionality remains intact.

[0056] In exemplary embodiments, the meal detection process **600** is implemented or otherwise performed by server **106**, for example, to detect or otherwise identify unannounced meals to facilitate generating GUI displays, notifications or alerts, or otherwise analyzing a patient's meal response. In some embodiments, the meal detection process **600** may be performed by the server **106** in response to receiving uploaded measurement data from the infusion device **102** and/or the sensing arrangement **104** to detect or otherwise identify a meal substantially in real-time. In other embodiments, the meal detection process **600** may be performed on a periodic basis independent of the infusion device **102** and/or the sensing arrangement **104**, for example, on a daily basis, a weekly basis, a monthly basis, or the like.

[0057] The meal detection process **600** begins by receiving or otherwise obtaining sensor glucose measurement values associated with a patient, determining sensor glucose measurement statistics associated with a preceding analysis interval, and determining a meal probability value for the preceding analysis interval (tasks **602**, **604**, **606**). In this regard, the server **106** obtains sensor glucose measurement values that are not accompanied by a meal indication, and calculates or otherwise determines sensor glucose measurement statistics based on the sensor glucose

measurement values corresponding to an analysis interval preceding the latest or most recent measurement value. For example, continuing the above example, the server **106** calculates sensor glucose measurement statistics based on the sensor glucose measurement values for the hour preceding the current sensor glucose measurement value being analyzed in a similar manner as described above (e.g., task **208**). The server **106** also determines a meal probability value based on an average of the meal probability intervals over the one hour interval preceding the current sensor glucose measurement value in a similar manner as described above (e.g., task **210**).

[0058] After determining sensor glucose measurement statistics and a meal probability value for an interval of sensor glucose measurement values to be analyzed, the meal detection process **600** obtains the meal detection model associated with the patient being analyzed and applies the detection model to the predictive subset of the sensor glucose measurement statistics and meal probability value to determine a metric indicative of whether the current analysis interval corresponds to a meal or non-meal segment (tasks **608**, **610**). In this regard, the server **106** implements the equation defined by the meal detection model associated with the identifier(s) for the patient being analyzed using current values for the predictive sensor glucose measurement statistic subset identified by the model and the current meal probability value (if indicated as predictive by the model) to obtain a result that quantifies or otherwise characterizes the likelihood of the patient having consumed a meal concurrently or contemporaneously to the current analysis interval. Alternatively, in instances where the patient does not have a patient-specific model assigned, the server **106** may utilize a patient group meal detection model corresponding to a patient group the patient is associated with.

[0059] Based on the output of the model, the meal detection process **600** determines whether the current analysis interval corresponds to a meal response by the patient being analyzed, and in response to detecting a meal, the meal detection process **600** provides indication of a meal associated with the current analysis interval (tasks **612**, **614**). In some embodiments, when the model output indicates the current analysis interval corresponds to a meal segment, the server **106** may analyze the patient's bolus data corresponding to the analysis interval to verify whether a meal bolus identified by the patient (e.g., a bolus accompanied by an input carbohydrate amount or other identification of a meal) was delivered during the analysis interval. When the patient's bolus data includes a bolus delivered during the analysis interval that is not marked as a meal bolus, the server **106** may flag, mark, or otherwise store an indication of a meal bolus in association with that bolus. Additionally, in some embodiments, where the model is capable of being utilized to classify a type of bolus delivered, the server **106** may also store an indication of a bolus type associated with that meal bolus. In some embodiments, in the absence of any bolus delivered during the analysis interval, the server **106** may store an indication of a bolus in association with one or more sensor glucose measurement values or the time period corresponding to the analysis interval to thereby flag, tag, or otherwise mark the analysis interval as a meal segment with a corresponding meal having been consumed by the patient. For example, in one embodiment, the server **106** calculates, estimates, or otherwise determines a time associated with consumption of the meal based on the sensor glucose measurements of the analysis interval and then creates or otherwise generates a meal indication associated with the patient having an associated timestamp corresponding to the estimated time of meal consumption. In this regard, the server **106** may update the patient's historical bolus data to indicate the meal associated with the analysis interval that was detected by the server **106** using the patient's meal detection model.

[0060] In response to detecting a meal, in addition to updating the patient's bolus history, the server **106** may update one or more GUI displays to reflect the meal, generate an alert or user notification based on the meal, or initiate or perform some other action in response to the meal. For example, if a monitoring GUI display depicting a graphical representation of the patient's sensor glucose is presented on a device **102**, **110** within the patient management system **100**, the server **106** may update the GUI display to include a marker for a meal bolus at a time corresponding to the analysis

interval, such as for example, at the time of a sensor glucose measurement value of the analysis interval (e.g., the time of the lowest sensor glucose measurement value) or another timestamp assigned to the meal bolus. Thus, a pattern guidance display depicting event patterns detected during operation of the infusion device **102** may depict an event pattern that is influenced by the detected meal and includes a graphical indication of the detected meal associated with a time of day corresponding to the analysis interval, as described in greater detail below in the context of FIG. 7. In yet other embodiments, the meal indication may be utilized to temporarily alter or influence operation of the infusion device **102** (e.g., to modify insulin delivery to account for the meal and/or bolus) as appropriate.

[0061] It should be noted that in general, an individual's blood sugar levels are generally influenced by food, exercise, medicine, sleep, and physiological stress (e.g., illness, mental stress, or the like). Accordingly, by virtue of the meal detection process **600** detecting a meal that could otherwise go unannounced or without indication by a user, other pattern detection algorithms can use this additional knowledge of meal consumption to better understand the patient's condition to provide recommendations or guidance for post-meal control, such as, for example, recommended medication regimen, eating habits or nutritional guidance, recommended proper medication types, or other behavioral profiling based on eating habits.

[0062] It should be noted that the meal detection process **600** may be incrementally performed on sensor glucose measurement values to classify or otherwise categorize as many non-overlapping segments of the patient history as meal or non-meal segments for use in analyzing the patient's meal response, validating the meal detection model, or the like. In this regard, when a meal bolus is manually-indicated or announced, the meal detection process **600** may still be performed to verify or otherwise confirm that the meal detection model generates an output indicative of a meal. For example, in one or more embodiment, the server **106** may dynamically update or track the performance of the model over time and detect when one or more performance metrics associated with the model fail to satisfy the validation criteria (e.g., task **218**). In this regard, when the model performance suffers, the patient modeling process **200** may be repeated to update the patient's meal detection model or reassign the patient to a population model as described above.

[0063] FIG. 7 depicts an exemplary embodiment of a snapshot GUI display **700** or report that may be presented on a display device associated with an electronic device **102**, **106**, **110** in the patient management system **100** in conjunction with the meal detection process **600** described above. The snapshot GUI display **700** includes a plurality of regions **702**, **704**, **706**, **708** that present information pertaining to past operation of the infusion device **102** to deliver insulin to regulate the glucose level of a diabetic patient. A header region **702** is presented at the top of the snapshot GUI display **700** and includes a graphical representation of a preceding time period of operation (e.g., November 3-November 6) associated with the snapshot GUI display **700** for which information is presented in the below regions **704**, **706**, **708**. A performance metric region **704** is presented below the header region **702** and includes graphical representations or other indicia of the values for various performance metrics calculated based on the historical measurement data for the patient's glucose level over the time period associated with the snapshot GUI display **700**, such as, for example, an average sensor glucose measurement value, an estimated A1C level, estimated percentages of the snapshot time period during which durations the sensor glucose measurement values were above, below or between one or more glucose threshold values, and the like.

[0064] The illustrated snapshot GUI display **700** includes a graph overlay region **708** presented at the bottom of the snapshot GUI display **700** that includes graphical representations of historical measurement data for the patient's glucose level over the snapshot time period with respect to time. In this regard, the graph overlay region **708** may include a line graph including a line associated with each day within the snapshot time period that depicts the patient's sensor glucose measurements values from that day with respect to time of day. Additionally, the graph overlay region **708** may include a line representative of the average of the patient's sensor glucose

measurements across the different days within the snapshot time period with respect to time of day. The illustrated graph overlay region **708** also includes a visually distinguishable overlay region that indicates a target range for the patient's sensor glucose measurement values. In exemplary embodiments, the graphical representation of the measurements for each different day or date depicted on the graph overlay region **708** is rendered with a unique color or other visually distinguishable characteristic relative to the graphical representations corresponding to other days or dates, with the meal markers **750** on that respective day or date also being rendered in the same color or visually distinguishable characteristic and placed on the line corresponding to that respective day or date.

[0065] Referring now to FIGS. **6-7**, in accordance with one or more embodiments, one or more of the meal markers **750** are generated in response to the meal detection process **600** detecting a meal segment within the historical measurement data for the patient's glucose level over the snapshot time period. For example, the meal detection process **600** may detect a meal segment corresponding to the period of 12:30 PM-1:30 PM of the patient's measurement data for November 5th, and in response, the meal detection process **600** may store or otherwise provide an indication of a meal associated with the sensor glucose measurement value corresponding to the start of that meal segment the absence of a meal indication within the patient's bolus history. Thus, even if the patient does not manually provide the meal indication, a corresponding meal marker **750** for that meal may still be presented on the snapshot GUI display **700**.

[0066] The illustrated snapshot GUI display **700** also includes a pattern detection region **706** that includes a plurality of pattern guidance displays **720, 730, 740**, where each pattern guidance display **720, 730, 740** corresponds to a respective pattern of events identified during the snapshot time period based on the patient's sensor glucose measurement values for the snapshot time period. In this regard, the historical sensor glucose measurement values are analyzed for different monitoring periods within the snapshot time period. Here, it is noted that the meal indications generated by the meal detection process **600** may be utilized to delineate or otherwise determine the times or portions of the day corresponding to monitoring periods referenced to meals. For example, the meal marker **750** identified based on the period of 12:30 PM-1:30 PM of the patient's measurement data for November 5th may be utilized to determine one or more of a starting time, an ending time, or a duration of a lunch time monitoring period in a manner that ensures the lunch time monitoring period includes or otherwise encompasses the meal detected by the meal detection process **600**. Thus, patterns for meal time monitoring period may be identified and analyzed even if the patient does not manually provide a meal indication for that particular meal. For example, a patient may routinely or habitually fail to manually indicate when he or she has consumed lunch due to the patient being preoccupied with other daily activities. By virtue of the patient-specific meal detection described above, lunch time event patterns for that patient may be better identified by tailoring the lunch time monitoring period to fit the patient's midday meal times. Thus, the duration of the lunch time monitoring period depicted by the lunch time monitoring period marker **728** described below encompasses the meal marker **750** detected by the meal detection process **600**.

[0067] Based on the subset of sensor glucose measurement values associated with times of day within a respective monitoring period, a pattern of one or more events is detected or otherwise identified within that monitoring period, such as, for example, a glucose variability event, a high glucose (or hyperglycemic) event, or a low glucose (or hypoglycemic) event. When a plurality of event patterns are identified, the identified event patterns are prioritized and filtered to limit the number of event patterns for display. For example, in one embodiment, the detected event patterns are prioritized primarily based on event type (e.g., from most significant to least significant) and secondarily based on the monitoring period associated with the respective event pattern, and then filtered to remove lower priority event patterns above a display threshold that limits the number of displayed event patterns.

[0068] For each remaining event pattern of the filtered prioritized list, a pattern guidance display

720, 730, 740 may be generated that includes graphical indicia of the event type and the monitoring period associated with the detected pattern, graphical indicia of the number, frequency, severity, or other characteristics of the events associated with the detected pattern, graphical indicia of potential causes or remedial actions for the detected events, and the like, which may also be prioritized or ordered according to their respective clinical relevance. Additionally, in exemplary embodiments, graphical indicia **728, 738, 748** of the detected event patterns are presented within the graph overlay region **708** in a manner that establishes an association between the detected event pattern, the time of day associated with its corresponding monitoring period, and its relative priority level. Thus, the graphical indicia **728, 738, 748** facilitate establishing an association between a respective subset of the historical measurement data presented within the graph overlay region **708** and a corresponding event pattern detected based on that subset of historical measurement data. For example, in the illustrated embodiment of FIG. 7, a marker **728** is presented overlying the graph overlay region **708** that includes an identifier that indicates the detected event pattern the marker **728** corresponds to (e.g., number 1 to indicate the highest priority event pattern **720**), and the marker **728** has a width or other dimension that encompasses or otherwise corresponds to the subset of the sensor glucose measurement values associated with the time of day corresponding to the monitoring period associated with the detected event pattern (e.g., the lunch time period).

[0069] FIG. 8 illustrates a computing device **800** including a display **833** suitable for presenting a snapshot GUI display **700** as part of a diabetes data management system in conjunction with the processes **200, 600** of FIGS. 2 and 6 described above. The diabetes data management system (DDMS) may be referred to as the Medtronic MiniMed CARELINK™ system or as a medical data management system (MDMS) in some embodiments. The DDMS may be housed on a server or a plurality of servers which a user or a health care professional may access via a communications network via the Internet or the World Wide Web. Some models of the DDMS, which is described as an MDMS, are described in U.S. Patent Application Publication Nos. 2006/0031094 and 2013/0338630, which is herein incorporated by reference in their entirety.

[0070] While description of embodiments are made in regard to monitoring medical or biological conditions for subjects having diabetes, the systems and processes herein are applicable to monitoring medical or biological conditions for cardiac subjects, cancer subjects, HIV subjects, subjects with other disease, infection, or controllable conditions, or various combinations thereof.

[0071] In various embodiments, the DDMS may be installed in a computing device in a health care provider's office, such as a doctor's office, a nurse's office, a clinic, an emergency room, an urgent care office. Health care providers may be reluctant to utilize a system where their confidential patient data is to be stored in a computing device such as a server on the Internet.

[0072] The DDMS may be installed on a computing device **800**. The computing device **800** may be coupled to a display **833**. In some embodiments, the computing device **800** may be in a physical device separate from the display (such as in a personal computer, a mini-computer, etc.) In some embodiments, the computing device **800** may be in a single physical enclosure or device with the display **833** such as a laptop where the display **833** is integrated into the computing device. In various embodiments, the computing device **800** hosting the DDMS may be, but is not limited to, a desktop computer, a laptop computer, a server, a network computer, a personal digital assistant (PDA), a portable telephone including computer functions, a pager with a large visible display, an insulin pump including a display, a glucose sensor including a display, a glucose meter including a display, and/or a combination insulin pump/glucose sensor having a display. The computing device may also be an insulin pump coupled to a display, a glucose meter coupled to a display, or a glucose sensor coupled to a display. The computing device **800** may also be a server located on the Internet that is accessible via a browser installed on a laptop computer, desktop computer, a network computer, or a PDA. The computing device **800** may also be a server located in a doctor's office that is accessible via a browser installed on a portable computing device, e.g., laptop, PDA, network computer, portable phone, which has wireless capabilities and can communicate via one of

the wireless communication protocols such as Bluetooth and IEEE 802.11 protocols.

[0073] In the embodiment shown in FIG. 8, the data management system **816** comprises a group of interrelated software modules or layers that specialize in different tasks. The system software includes a device communication layer **824**, a data parsing layer **826**, a database layer **828**, database storage devices **829**, a reporting layer **830**, a graph display layer **831**, and a user interface layer **832**. The diabetes data management system may communicate with a plurality of subject support devices **812**, two of which are illustrated in FIG. 8. Although the different reference numerals refer to a number of layers, (e.g., a device communication layer, a data parsing layer, a database layer), each layer may include a single software module or a plurality of software modules. For example, the device communications layer **824** may include a number of interacting software modules, libraries, etc. In some embodiments, the data management system **816** may be installed onto a non-volatile storage area (memory such as flash memory, hard disk, removable hard, DVD-RW, CD-RW) of the computing device **800**. If the data management system **816** is selected or initiated, the system **816** may be loaded into a volatile storage (memory such as DRAM, SRAM, RAM, DDRAM) for execution.

[0074] The device communication layer **824** is responsible for interfacing with at least one, and, in further embodiments, to a plurality of different types of subject support devices **812**, such as, for example, blood glucose meters, glucose sensors/monitors, or an infusion pump. In one embodiment, the device communication layer **824** may be configured to communicate with a single type of subject support device **812**. However, in more comprehensive embodiments, the device communication layer **824** is configured to communicate with multiple different types of subject support devices **812**, such as devices made from multiple different manufacturers, multiple different models from a particular manufacturer and/or multiple different devices that provide different functions (such as infusion functions, sensing functions, metering functions, communication functions, user interface functions, or combinations thereof). By providing an ability to interface with multiple different types of subject support devices **812**, the diabetes data management system **816** may collect data from a significantly greater number of discrete sources. Such embodiments may provide expanded and improved data analysis capabilities by including a greater number of subjects and groups of subjects in statistical or other forms of analysis that can benefit from larger amounts of sample data and/or greater diversity in sample data, and, thereby, improve capabilities of determining appropriate treatment parameters, diagnostics, or the like.

[0075] The device communication layer **824** allows the DDMS **816** to receive information from and transmit information to or from each subject support device **812** in the system **816**. Depending upon the embodiment and context of use, the type of information that may be communicated between the system **816** and device **812** may include, but is not limited to, data, programs, updated software, education materials, warning messages, notifications, device settings, therapy parameters, or the like. The device communication layer **824** may include suitable routines for detecting the type of subject support device **812** in communication with the system **816** and implementing appropriate communication protocols for that type of device **812**. Alternatively or in addition, the subject support device **812** may communicate information in packets or other data arrangements, where the communication includes a preamble or other portion that includes device identification information for identifying the type of the subject support device. Alternatively, or in addition, the subject support device **812** may include suitable user-operable interfaces for allowing a user to enter information, such as by selecting an optional icon or text or other device identifier, that corresponds to the type of subject support device used by that user. Such information may be communicated to the system **816**, through a network connection. In yet further embodiments, the system **816** may detect the type of subject support device **812** it is communicating with and then may send a message requiring the user to verify that the system **816** properly detected the type of subject support device being used by the user. For systems **816** that are capable of communicating with multiple different types of subject support devices **812**, the device communication layer **824**

may be capable of implementing multiple different communication protocols and selects a protocol that is appropriate for the detected type of subject support device.

[0076] The data-parsing layer **826** is responsible for validating the integrity of device data received and for inputting it correctly into a database **829**. A cyclic redundancy check CRC process for checking the integrity of the received data may be employed. Alternatively, or in addition, data may be received in packets or other data arrangements, where preambles or other portions of the data include device type identification information. Such preambles or other portions of the received data may further include device serial numbers or other identification information that may be used for validating the authenticity of the received information. In such embodiments, the system **816** may compare received identification information with pre-stored information to evaluate whether the received information is from a valid source.

[0077] The database layer **828** may include a centralized database repository that is responsible for warehousing and archiving stored data in an organized format for later access, and retrieval. The database layer **828** operates with one or more data storage device(s) **829** suitable for storing and providing access to data in the manner described herein. Such data storage device(s) **829** may comprise, for example, one or more hard discs, optical discs, tapes, digital libraries or other suitable digital or analog storage media and associated drive devices, drive arrays or the like.

[0078] Data may be stored and archived for various purposes, depending upon the embodiment and environment of use. Information regarding specific subjects and patient support devices may be stored and archived and made available to those specific subjects, their authorized healthcare providers and/or authorized healthcare payor entities for analyzing the subject's condition. Also, certain information regarding groups of subjects or groups of subject support devices may be made available more generally for healthcare providers, subjects, personnel of the entity administering the system **816** or other entities, for analyzing group data or other forms of conglomerate data.

[0079] Embodiments of the database layer **828** and other components of the system **816** may employ suitable data security measures for securing personal medical information of subjects, while also allowing non-personal medical information to be more generally available for analysis. Embodiments may be configured for compliance with suitable government regulations, industry standards, policies or the like, including, but not limited to the Health Insurance Portability and Accountability Act of 1996 (HIPAA).

[0080] The database layer **828** may be configured to limit access of each user to types of information pre-authorized for that user. For example, a subject may be allowed access to his or her individual medical information (with individual identifiers) stored by the database layer **828**, but not allowed access to other subject's individual medical information (with individual identifiers). Similarly, a subject's authorized healthcare provider or payor entity may be provided access to some or all of the subject's individual medical information (with individual identifiers) stored by the database layer **828**, but not allowed access to another individual's personal information. Also, an operator or administrator-user (on a separate computer communicating with the computing device **800**) may be provided access to some or all subject information, depending upon the role of the operator or administrator. On the other hand, a subject, healthcare provider, operator, administrator or other entity, may be authorized to access general information of unidentified individuals, groups or conglomerates (without individual identifiers) stored by the database layer **828** in the data storage devices **829**.

[0081] In embodiments of the subject matter described herein, the database layer **828** may store patient-specific event detection models, population group detection models, and individual patient preference profiles. In the database layer **828**, for example, each user may store information regarding specific parameters that correspond to the user. Illustratively, these parameters could include target blood glucose or sensor glucose levels, what type of equipment the users utilize (insulin pump, glucose sensor, blood glucose meter, etc.) and could be stored in a record, a file, or a memory location in the data storage device(s) **829** in the database layer. The preference profiles

may include various threshold values, monitoring period values, prioritization criteria, filtering criteria, and/or other user-specific values for parameters utilized to generate a snapshot GUI display, such as snapshot GUI display **700**, on the display **833** or a support device **812** in a personalized or patient-specific manner. Additionally, data or information defining the meal detection or other event detection models associated with a particular individual may also be stored in a record, a file, or a memory location associated with that patient in the data storage device(s) **829** in the database layer.

[0082] The DDMS **816** may measure, analyze, and track either blood glucose (BG) or sensor glucose (SG) readings for a user. In exemplary embodiments, the medical data management system may measure, track, or analyze both BG and SG readings for the user. Accordingly, although certain reports may mention or illustrate BG or SG only, the reports may monitor and display results for the other one of the glucose readings or for both of the glucose readings.

[0083] The reporting layer **830** may include a report wizard program that pulls data from selected locations in the database **829** and generates report information from the desired parameters of interest. The reporting layer **830** may be configured to generate multiple different types of reports, each having different information and/or showing information in different formats (arrangements or styles), where the type of report may be selectable by the user. A plurality of pre-set types of report (with pre-defined types of content and format) may be available and selectable by a user. At least some of the pre-set types of reports may be common, industry standard report types with which many healthcare providers should be familiar. In exemplary embodiments described herein, the reporting layer **830** also facilitates generation of a snapshot report including a snapshot GUI display, such as snapshot GUI display **700** of FIG. 7.

[0084] In some embodiments, the database layer **828** may calculate values for various medical information that is to be displayed on the reports generated by the report or reporting layer **830**. For example, the database layer **828**, may calculate average blood glucose or sensor glucose readings for specified timeframes. In some embodiments, the reporting layer **830** may calculate values for medical or physical information that is to be displayed on the reports. For example, a user may select parameters which are then utilized by the reporting layer **830** to generate medical information values corresponding to the selected parameters. In other embodiments, the user may select a parameter profile that previously existed in the database layer **828**.

[0085] Alternatively, or in addition, the report wizard may allow a user to design a custom type of report. For example, the report wizard may allow a user to define and input parameters (such as parameters specifying the type of content data, the time period of such data, the format of the report, or the like) and may select data from the database and arrange the data in a printable or displayable arrangement, based on the user-defined parameters. In further embodiments, the report wizard may interface with or provide data for use by other programs that may be available to users, such as common report generating, formatting or statistical analysis programs. In this manner, users may import data from the system **816** into further reporting tools familiar to the user. The reporting layer **830** may generate reports in displayable form to allow a user to view reports on a standard display device, printable form to allow a user to print reports on standard printers, or other suitable forms for access by a user. Embodiments may operate with conventional file format schemes for simplifying storing, printing and transmitting functions, including, but not limited to PDF, JPEG, or the like. Illustratively, a user may select a type of report and parameters for the report and the reporting layer **830** may create the report in a PDF format. A PDF plug-in may be initiated to help create the report and also to allow the user to view the report. Under these operating conditions, the user may print the report utilizing the PDF plug-in. In certain embodiments in which security measures are implemented, for example, to meet government regulations, industry standards or policies that restrict communication of subject's personal information, some or all reports may be generated in a form (or with suitable software controls) to inhibit printing, or electronic transfer (such as a non-printable and/or non-capable format). In yet further embodiments, the system **816**

may allow a user generating a report to designate the report as non-printable and/or non-transferable, whereby the system **816** will provide the report in a form that inhibits printing and/or electronic transfer.

[0086] The reporting layer **830** may transfer selected reports to the graph display layer **831**. The graph display layer **831** receives information regarding the selected reports and converts the data into a format that can be displayed or shown on a display **833**.

[0087] In various embodiments, the reporting layer **830** may store a number of the user's parameters. Illustratively, the reporting layer **830** may store the type of carbohydrate units, a blood glucose movement or sensor glucose reading, a carbohydrate conversion factor, and timeframes for specific types of reports. These examples are meant to be illustrative and not limiting.

[0088] Data analysis and presentations of the reported information may be employed to develop and support diagnostic and therapeutic parameters. Where information on the report relates to an individual subject, the diagnostic and therapeutic parameters may be used to assess the health status and relative well-being of that subject, assess the subject's compliance to a therapy, as well as to develop or modify treatment for the subject and assess the subject's behaviors that affect his/her therapy. Where information on the report relates to groups of subjects or conglomerates of data, the diagnostic and therapeutic parameters may be used to assess the health status and relative well-being of groups of subjects with similar medical conditions, such as, but not limited to, diabetic subjects, cardiac subjects, diabetic subjects having a particular type of diabetes or cardiac condition, subjects of a particular age, sex or other demographic group, subjects with conditions that influence therapeutic decisions such as but not limited to pregnancy, obesity, hypoglycemic unawareness, learning disorders, limited ability to care for self, various levels of insulin resistance, combinations thereof, or the like.

[0089] The user interface layer **832** supports interactions with the end user, for example, for user login and data access, software navigation, data input, user selection of desired report types and the display of selected information. Users may also input parameters to be utilized in the selected reports via the user interface layer **832**. Examples of users include but are not limited to: healthcare providers, healthcare payer entities, system operators or administrators, researchers, business entities, healthcare institutions and organizations, or the like, depending upon the service being provided by the system and depending upon the embodiment. More comprehensive embodiments are capable of interacting with some or all of the above-noted types of users, wherein different types of users have access to different services or data or different levels of services or data.

[0090] In an example embodiment, the user interface layer **832** provides one or more websites accessible by users on the Internet. The user interface layer may include or operate with at least one (or multiple) suitable network server(s) to provide the website(s) over the Internet and to allow access, world-wide, from Internet-connected computers using standard Internet browser software. The website(s) may be accessed by various types of users, including but not limited to subjects, healthcare providers, researchers, business entities, healthcare institutions and organizations, payor entities, pharmaceutical partners or other sources of pharmaceuticals or medical equipment, and/or support personnel or other personnel running the system **816**, depending upon the embodiment of use.

[0091] In another example embodiment, where the DDMS **816** is located on one computing device **800**, the user interface layer **832** provides a number of menus to the user to navigate through the DDMS. These menus may be created utilizing any menu format, including but not limited to HTML, XML, or Active Server pages. A user may access the DDMS **816** to perform one or more of a variety of tasks, such as accessing general information made available on a website to all subjects or groups of subjects. The user interface layer **832** of the DDMS **816** may allow a user to access specific information or to generate reports regarding that subject's medical condition or that subject's medical device(s) **812**, to transfer data or other information from that subject's support device(s) **812** to the system **816**, to transfer data, programs, program updates or other information

from the system **816** to the subject's support device(s) **812**, to manually enter information into the system **816**, to engage in a remote consultation exchange with a healthcare provider, or to modify the custom settings in a subject's supported device and/or in a subject's DDMS/MDMS data file. [0092] The system **816** may provide access to different optional resources or activities (including accessing different information items and services) to different users and to different types or groups of users, such that each user may have a customized experience and/or each type or group of user (e.g., all users, diabetic users, cardio users, healthcare provider-user or payor-user, or the like) may have a different set of information items or services available on the system. The system **816** may include or employ one or more suitable resource provisioning program or system for allocating appropriate resources to each user or type of user, based on a pre-defined authorization plan. Resource provisioning systems are well known in connection with provisioning of electronic office resources (email, software programs under license, sensitive data, etc.) in an office environment, for example, in a local area network LAN for an office, company or firm. In one example embodiment, such resource provisioning systems is adapted to control access to medical information and services on the DDMS **816**, based on the type of user and/or the identity of the user.

[0093] Upon entering successful verification of the user's identification information and password, the user may be provided access to secure, personalized information stored on the DDMS **816**. For example, the user may be provided access to a secure, personalized location in the DDMS **816** which has been assigned to the subject. This personalized location may be referred to as a personalized screen, a home screen, a home menu, a personalized page, etc. The personalized location may provide a personalized home screen to the subject, including selectable icons or menu items for selecting optional activities, including, for example, an option to transfer device data from a subject's supported device **812** to the system **816**, manually enter additional data into the system **816**, modify the subject's custom settings, and/or view and print reports. Reports may include data specific to the subject's condition, including but not limited to, data obtained from the subject's subject support device(s) **812**, data manually entered, data from medical libraries or other networked therapy management systems, data from the subjects or groups of subjects, or the like. Where the reports include subject-specific information and subject identification information, the reports may be generated from some or all subject data stored in a secure storage area (e.g., storage devices **829**) employed by the database layer **828**.

[0094] The user may select an option to transfer (send) device data to the medical data management system **816**. If the system **816** receives a user's request to transfer device data to the system, the system **816** may provide the user with step-by-step instructions on how to transfer data from the subject's supported device(s) **812**. For example, the DDMS **816** may have a plurality of different stored instruction sets for instructing users how to download data from different types of subject support devices, where each instruction set relates to a particular type of subject supported device (e.g., pump, sensor, meter, or the like), a particular manufacturer's version of a type of subject support device, or the like. Registration information received from the user during registration may include information regarding the type of subject support device(s) **812** used by the subject. The system **816** employs that information to select the stored instruction set(s) associated with the particular subject's support device(s) **812** for display to the user.

[0095] Other activities or resources available to the user on the system **816** may include an option for manually entering information to the DDMS/MDMS **816**. For example, from the user's personalized menu or location, the user may select an option to manually enter additional information into the system **816**.

[0096] Further optional activities or resources may be available to the user on the DDMS **816**. For example, from the user's personalized menu, the user may select an option to receive data, software, software updates, treatment recommendations or other information from the system **816** on the subject's support device(s) **812**. If the system **816** receives a request from a user to receive

data, software, software updates, treatment recommendations or other information, the system **816** may provide the user with a list or other arrangement of multiple selectable icons or other indicia representing available data, software, software updates or other information available to the user. [0097] Yet further optional activities or resources may be available to the user on the medical data management system **816** including, for example, an option for the user to customize or otherwise further personalize the user's personalized location or menu. In particular, from the user's personalized location, the user may select an option to customize parameters for the user. In addition, the user may create profiles of customizable parameters. When the system **816** receives such a request from a user, the system **816** may provide the user with a list or other arrangement of multiple selectable icons or other indicia representing parameters that may be modified to accommodate the user's preferences. When a user selects one or more of the icons or other indicia, the system **816** may receive the user's request and makes the requested modification.

[0098] FIG. **9** depicts one exemplary embodiment of an infusion system **900** that includes, without limitation, a fluid infusion device (or infusion pump) **902**, a sensing arrangement **904**, a command control device (CCD) **906**, and a computer **908**, which could be realized as any one of the computing devices **106**, **110**, **800**, **812** described above. The components of an infusion system **900** may be realized using different platforms, designs, and configurations, and the embodiment shown in FIG. **9** is not exhaustive or limiting. In practice, the infusion device **902** and the sensing arrangement **904** are secured at desired locations on the body of a user (or patient), as illustrated in FIG. **9**. In this regard, the locations at which the infusion device **902** and the sensing arrangement **904** are secured to the body of the user in FIG. **9** are provided only as a representative, non-limiting, example. The elements of the infusion system **900** may be similar to those described in U.S. Pat. No. 8,674,288, the subject matter of which is hereby incorporated by reference in its entirety.

[0099] In the illustrated embodiment of FIG. **9**, the infusion device **902** is designed as a portable medical device suitable for infusing a fluid, a liquid, a gel, or other agent into the body of a user. In exemplary embodiments, the infused fluid is insulin, although many other fluids may be administered through infusion such as, but not limited to, HIV drugs, drugs to treat pulmonary hypertension, iron chelation drugs, pain medications, anti-cancer treatments, medications, vitamins, hormones, or the like. In some embodiments, the fluid may include a nutritional supplement, a dye, a tracing medium, a saline medium, a hydration medium, or the like.

[0100] The sensing arrangement **904** generally represents the components of the infusion system **900** configured to sense, detect, measure or otherwise quantify a condition of the user, and may include a sensor, a monitor, or the like, for providing data indicative of the condition that is sensed, detected, measured or otherwise monitored by the sensing arrangement. In this regard, the sensing arrangement **904** may include electronics and enzymes reactive to a biological or physiological condition of the user, such as a blood glucose level, or the like, and provide data indicative of the blood glucose level to the infusion device **902**, the CCD **906** and/or the computer **908**. For example, the infusion device **902**, the CCD **906** and/or the computer **908** may include a display for presenting information or data to the user based on the sensor data received from the sensing arrangement **904**, such as, for example, a current glucose level of the user, a graph or chart of the user's glucose level versus time, device status indicators, alert messages, or the like. In other embodiments, the infusion device **902**, the CCD **906** and/or the computer **908** may include electronics and software that are configured to analyze sensor data and operate the infusion device **902** to deliver fluid to the body of the user based on the sensor data and/or preprogrammed delivery routines. Thus, in exemplary embodiments, one or more of the infusion device **902**, the sensing arrangement **904**, the CCD **906**, and/or the computer **908** includes a transmitter, a receiver, and/or other transceiver electronics that allow for communication with other components of the infusion system **900**, so that the sensing arrangement **904** may transmit sensor data or monitor data to one or more of the infusion device **902**, the CCD **906** and/or the computer **908**.

[0101] Still referring to FIG. 9, in various embodiments, the sensing arrangement **904** may be secured to the body of the user or embedded in the body of the user at a location that is remote from the location at which the infusion device **902** is secured to the body of the user. In various other embodiments, the sensing arrangement **904** may be incorporated within the infusion device **902**. In other embodiments, the sensing arrangement **904** may be separate and apart from the infusion device **902**, and may be, for example, part of the CCD **906**. In such embodiments, the sensing arrangement **904** may be configured to receive a biological sample, analyte, or the like, to measure a condition of the user.

[0102] In various embodiments, the CCD **906** and/or the computer **908** may include electronics and other components configured to perform processing, delivery routine storage, and to control the infusion device **902** in a manner that is influenced by sensor data measured by and/or received from the sensing arrangement **904**. By including control functions in the CCD **906** and/or the computer **908**, the infusion device **902** may be made with more simplified electronics. However, in other embodiments, the infusion device **902** may include all control functions, and may operate without the CCD **906** and/or the computer **908**. In various embodiments, the CCD **906** may be a portable electronic device. In addition, in various embodiments, the infusion device **902** and/or the sensing arrangement **904** may be configured to transmit data to the CCD **906** and/or the computer **908** for display or processing of the data by the CCD **906** and/or the computer **908**.

[0103] In some embodiments, the CCD **906** and/or the computer **908** may provide information to the user that facilitates the user's subsequent use of the infusion device **902**. For example, the CCD **906** may provide information to the user to allow the user to determine the rate or dose of medication to be administered into the user's body. In other embodiments, the CCD **906** may provide information to the infusion device **902** to autonomously control the rate or dose of medication administered into the body of the user. In some embodiments, the sensing arrangement **904** may be integrated into the CCD **906**. Such embodiments may allow the user to monitor a condition by providing, for example, a sample of his or her blood to the sensing arrangement **904** to assess his or her condition. In some embodiments, the sensing arrangement **904** and the CCD **906** may be used for determining glucose levels in the blood and/or body fluids of the user without the use of, or necessity of, a wire or cable connection between the infusion device **902** and the sensing arrangement **904** and/or the CCD **906**.

[0104] In one or more exemplary embodiments, the sensing arrangement **904** and/or the infusion device **902** are cooperatively configured to utilize a closed-loop system for delivering fluid to the user. Examples of sensing devices and/or infusion pumps utilizing closed-loop systems may be found at, but are not limited to, the following U.S. Pat. Nos. 6,088,608, 6,119,028, 6,589,229, 6,740,072, 6,827,702, 7,323,142, and 7,402,153, all of which are incorporated herein by reference in their entirety. In such embodiments, the sensing arrangement **904** is configured to sense or measure a condition of the user, such as, blood glucose level or the like. The infusion device **902** is configured to deliver fluid in response to the condition sensed by the sensing arrangement **904**. In turn, the sensing arrangement **904** continues to sense or otherwise quantify a current condition of the user, thereby allowing the infusion device **902** to deliver fluid continuously in response to the condition currently (or most recently) sensed by the sensing arrangement **904** indefinitely. In some embodiments, the sensing arrangement **904** and/or the infusion device **902** may be configured to utilize the closed-loop system only for a portion of the day, for example only when the user is asleep or awake.

[0105] FIGS. **10-12** depict one exemplary embodiment of a fluid infusion device **1000** (or alternatively, infusion pump) suitable for use in an infusion system, such as, for example, as infusion device **902** in the infusion system **900** of FIG. **9** or as infusion device **102** in the patient management system **100** of FIG. **1**. The fluid infusion device **1000** is a portable medical device designed to be carried or worn by a patient (or user), and the fluid infusion device **1000** may leverage any number of conventional features, components, elements, and characteristics of

existing fluid infusion devices, such as, for example, some of the features, components, elements, and/or characteristics described in U.S. Pat. Nos. 6,485,465 and 7,621,893. It should be appreciated that FIGS. **10-12** depict some aspects of the infusion device **1000** in a simplified manner; in practice, the infusion device **1000** could include additional elements, features, or components that are not shown or described in detail herein.

[0106] As best illustrated in FIGS. **10-11**, the illustrated embodiment of the fluid infusion device **1000** includes a housing **1002** adapted to receive a fluid-containing reservoir **1005**. An opening **1020** in the housing **1002** accommodates a fitting **1023** (or cap) for the reservoir **1005**, with the fitting **1023** being configured to mate or otherwise interface with tubing **1021** of an infusion set **1025** that provides a fluid path to/from the body of the user. In this manner, fluid communication from the interior of the reservoir **1005** to the user is established via the tubing **1021**. The illustrated fluid infusion device **1000** includes a human-machine interface (HMI) **1030** (or user interface) that includes elements **1032**, **1034** that can be manipulated by the user to administer a bolus of fluid (e.g., insulin), to change therapy settings, to change user preferences, to select display features, and the like. The infusion device also includes a display element **1026**, such as a liquid crystal display (LCD) or another suitable display element, that can be used to present various types of information or data to the user, such as, without limitation: the current glucose level of the patient; the time; a graph or chart of the patient's glucose level versus time; device status indicators; etc.

[0107] The housing **1002** is formed from a substantially rigid material having a hollow interior **1014** adapted to allow an electronics assembly **1004**, a sliding member (or slide) **1006**, a drive system **1008**, a sensor assembly **1010**, and a drive system capping member **1012** to be disposed therein in addition to the reservoir **1005**, with the contents of the housing **1002** being enclosed by a housing capping member **1016**. The opening **1020**, the slide **1006**, and the drive system **1008** are coaxially aligned in an axial direction (indicated by arrow **1018**), whereby the drive system **1008** facilitates linear displacement of the slide **1006** in the axial direction **1018** to dispense fluid from the reservoir **1005** (after the reservoir **1005** has been inserted into opening **1020**), with the sensor assembly **1010** being configured to measure axial forces (e.g., forces aligned with the axial direction **1018**) exerted on the sensor assembly **1010** responsive to operating the drive system **1008** to displace the slide **1006**. In various embodiments, the sensor assembly **1010** may be utilized to detect one or more of the following: an occlusion in a fluid path that slows, prevents, or otherwise degrades fluid delivery from the reservoir **1005** to a user's body; when the reservoir **1005** is empty; when the slide **1006** is properly seated with the reservoir **1005**; when a fluid dose has been delivered; when the infusion pump **1000** is subjected to shock or vibration; when the infusion pump **1000** requires maintenance.

[0108] Depending on the embodiment, the fluid-containing reservoir **1005** may be realized as a syringe, a vial, a cartridge, a bag, or the like. In certain embodiments, the infused fluid is insulin, although many other fluids may be administered through infusion such as, but not limited to, HIV drugs, drugs to treat pulmonary hypertension, iron chelation drugs, pain medications, anti-cancer treatments, medications, vitamins, hormones, or the like. As best illustrated in FIGS. **11-12**, the reservoir **1005** typically includes a reservoir barrel **1019** that contains the fluid and is concentrically and/or coaxially aligned with the slide **1006** (e.g., in the axial direction **1018**) when the reservoir **1005** is inserted into the infusion pump **1000**. The end of the reservoir **1005** proximate the opening **1020** may include or otherwise mate with the fitting **1023**, which secures the reservoir **1005** in the housing **1002** and prevents displacement of the reservoir **1005** in the axial direction **1018** with respect to the housing **1002** after the reservoir **1005** is inserted into the housing **1002**. As described above, the fitting **1023** extends from (or through) the opening **1020** of the housing **1002** and mates with tubing **1021** to establish fluid communication from the interior of the reservoir **1005** (e.g., reservoir barrel **1019**) to the user via the tubing **1021** and infusion set **1025**. The opposing end of the reservoir **1005** proximate the slide **1006** includes a plunger **1017** (or stopper) positioned to push fluid from inside the barrel **1019** of the reservoir **1005** along a fluid path through tubing **1021** to a

user. The slide **1006** is configured to mechanically couple or otherwise engage with the plunger **1017**, thereby becoming seated with the plunger **1017** and/or reservoir **1005**. Fluid is forced from the reservoir **1005** via tubing **1021** as the drive system **1008** is operated to displace the slide **1006** in the axial direction **1018** toward the opening **1020** in the housing **1002**.

[0109] In the illustrated embodiment of FIGS. **11-12**, the drive system **1008** includes a motor assembly **1007** and a drive screw **1009**. The motor assembly **1007** includes a motor that is coupled to drive train components of the drive system **1008** that are configured to convert rotational motor motion to a translational displacement of the slide **1006** in the axial direction **1018**, and thereby engaging and displacing the plunger **1017** of the reservoir **1005** in the axial direction **1018**. In some embodiments, the motor assembly **1007** may also be powered to translate the slide **1006** in the opposing direction (e.g., the direction opposite direction **1018**) to retract and/or detach from the reservoir **1005** to allow the reservoir **1005** to be replaced. In exemplary embodiments, the motor assembly **1007** includes a brushless DC (BLDC) motor having one or more permanent magnets mounted, affixed, or otherwise disposed on its rotor. However, the subject matter described herein is not necessarily limited to use with BLDC motors, and in alternative embodiments, the motor may be realized as a solenoid motor, an AC motor, a stepper motor, a piezoelectric caterpillar drive, a shape memory actuator drive, an electrochemical gas cell, a thermally driven gas cell, a bimetallic actuator, or the like. The drive train components may comprise one or more lead screws, cams, ratchets, jacks, pulleys, pawls, clamps, gears, nuts, slides, bearings, levers, beams, stoppers, plungers, sliders, brackets, guides, bearings, supports, bellows, caps, diaphragms, bags, heaters, or the like. In this regard, although the illustrated embodiment of the infusion pump utilizes a coaxially aligned drive train, the motor could be arranged in an offset or otherwise non-coaxial manner, relative to the longitudinal axis of the reservoir **1005**.

[0110] As best shown in FIG. **12**, the drive screw **1009** mates with threads **1202** internal to the slide **1006**. When the motor assembly **1007** is powered and operated, the drive screw **1009** rotates, and the slide **1006** is forced to translate in the axial direction **1018**. In an exemplary embodiment, the infusion pump **1000** includes a sleeve **1011** to prevent the slide **1006** from rotating when the drive screw **1009** of the drive system **1008** rotates. Thus, rotation of the drive screw **1009** causes the slide **1006** to extend or retract relative to the drive motor assembly **1007**. When the fluid infusion device is assembled and operational, the slide **1006** contacts the plunger **1017** to engage the reservoir **1005** and control delivery of fluid from the infusion pump **1000**. In an exemplary embodiment, the shoulder portion **1015** of the slide **1006** contacts or otherwise engages the plunger **1017** to displace the plunger **1017** in the axial direction **1018**. In alternative embodiments, the slide **1006** may include a threaded tip **1013** capable of being detachably engaged with internal threads **1204** on the plunger **1017** of the reservoir **1005**, as described in detail in U.S. Pat. Nos. 6,248,093 and 6,485,465, which are incorporated by reference herein.

[0111] As illustrated in FIG. **11**, the electronics assembly **1004** includes control electronics **1024** coupled to the display element **1026**, with the housing **1002** including a transparent window portion **1028** that is aligned with the display element **1026** to allow the display **1026** to be viewed by the user when the electronics assembly **1004** is disposed within the interior **1014** of the housing **1002**. The control electronics **1024** generally represent the hardware, firmware, processing logic and/or software (or combinations thereof) configured to control operation of the motor assembly **1007** and/or drive system **1008**, as described in greater detail below in the context of FIG. **13**. Whether such functionality is implemented as hardware, firmware, a state machine, or software depends upon the particular application and design constraints imposed on the embodiment. Those familiar with the concepts described here may implement such functionality in a suitable manner for each particular application, but such implementation decisions should not be interpreted as being restrictive or limiting. In an exemplary embodiment, the control electronics **1024** includes one or more programmable controllers that may be programmed to control operation of the infusion pump **1000**.

[0112] The motor assembly **1007** includes one or more electrical leads **1036** adapted to be electrically coupled to the electronics assembly **1004** to establish communication between the control electronics **1024** and the motor assembly **1007**. In response to command signals from the control electronics **1024** that operate a motor driver (e.g., a power converter) to regulate the amount of power supplied to the motor from a power supply, the motor actuates the drive train components of the drive system **1008** to displace the slide **1006** in the axial direction **1018** to force fluid from the reservoir **1005** along a fluid path (including tubing **1021** and an infusion set), thereby administering doses of the fluid contained in the reservoir **1005** into the user's body. Preferably, the power supply is realized one or more batteries contained within the housing **1002**. Alternatively, the power supply may be a solar panel, capacitor, AC or DC power supplied through a power cord, or the like. In some embodiments, the control electronics **1024** may operate the motor of the motor assembly **1007** and/or drive system **1008** in a stepwise manner, typically on an intermittent basis; to administer discrete precise doses of the fluid to the user according to programmed delivery profiles.

[0113] Referring to FIGS. **10-12**, as described above, the user interface **1030** includes HMI elements, such as buttons **1032** and a directional pad **1034**, that are formed on a graphic keypad overlay **1031** that overlies a keypad assembly **1033**, which includes features corresponding to the buttons **1032**, directional pad **1034** or other user interface items indicated by the graphic keypad overlay **1031**. When assembled, the keypad assembly **1033** is coupled to the control electronics **1024**, thereby allowing the HMI elements **1032**, **1034** to be manipulated by the user to interact with the control electronics **1024** and control operation of the infusion pump **1000**, for example, to administer a bolus of insulin, to change therapy settings, to change user preferences, to select display features, to set or disable alarms and reminders, and the like. In this regard, the control electronics **1024** maintains and/or provides information to the display **1026** regarding program parameters, delivery profiles, pump operation, alarms, warnings, statuses, or the like, which may be adjusted using the HMI elements **1032**, **1034**. In various embodiments, the HMI elements **1032**, **1034** may be realized as physical objects (e.g., buttons, knobs, joysticks, and the like) or virtual objects (e.g., using touch-sensing and/or proximity-sensing technologies). For example, in some embodiments, the display **1026** may be realized as a touch screen or touch-sensitive display, and in such embodiments, the features and/or functionality of the HMI elements **1032**, **1034** may be integrated into the display **1026** and the HMI **1030** may not be present. In some embodiments, the electronics assembly **1004** may also include alert generating elements coupled to the control electronics **1024** and suitably configured to generate one or more types of feedback, such as, without limitation: audible feedback; visual feedback; haptic (physical) feedback; or the like.

[0114] Referring to FIGS. **11-12**, in accordance with one or more embodiments, the sensor assembly **1010** includes a back plate structure **1050** and a loading element **1060**. The loading element **1060** is disposed between the capping member **1012** and a beam structure **1070** that includes one or more beams having sensing elements disposed thereon that are influenced by compressive force applied to the sensor assembly **1010** that deflects the one or more beams, as described in greater detail in U.S. Pat. No. 8,474,332, which is incorporated by reference herein. In exemplary embodiments, the back plate structure **1050** is affixed, adhered, mounted, or otherwise mechanically coupled to the bottom surface **1038** of the drive system **1008** such that the back plate structure **1050** resides between the bottom surface **1038** of the drive system **1008** and the housing cap **1016**. The drive system capping member **1012** is contoured to accommodate and conform to the bottom of the sensor assembly **1010** and the drive system **1008**. The drive system capping member **1012** may be affixed to the interior of the housing **1002** to prevent displacement of the sensor assembly **1010** in the direction opposite the direction of force provided by the drive system **1008** (e.g., the direction opposite direction **1018**). Thus, the sensor assembly **1010** is positioned between the motor assembly **1007** and secured by the capping member **1012**, which prevents displacement of the sensor assembly **1010** in a downward direction opposite the direction of arrow **1018**, such that the sensor assembly **1010** is subjected to a reactionary compressive force when the

drive system **1008** and/or motor assembly **1007** is operated to displace the slide **1006** in the axial direction **1018** in opposition to the fluid pressure in the reservoir **1005**. Under normal operating conditions, the compressive force applied to the sensor assembly **1010** is correlated with the fluid pressure in the reservoir **1005**. As shown, electrical leads **1040** are adapted to electrically couple the sensing elements of the sensor assembly **1010** to the electronics assembly **1004** to establish communication to the control electronics **1024**, wherein the control electronics **1024** are configured to measure, receive, or otherwise obtain electrical signals from the sensing elements of the sensor assembly **1010** that are indicative of the force applied by the drive system **1008** in the axial direction **1018**.

[0115] FIG. **13** depicts an exemplary embodiment of a control system **1300** suitable for use with an infusion device **1302**, such as any one of the infusion devices **102**, **902**, **1000** described above. The control system **1300** is capable of controlling or otherwise regulating a physiological condition in the body **1301** of a user to a desired (or target) value or otherwise maintain the condition within a range of acceptable values in an automated or autonomous manner. In one or more exemplary embodiments, the condition being regulated is sensed, detected, measured or otherwise quantified by a sensing arrangement **1304** (e.g., sensing arrangement **904**) communicatively coupled to the infusion device **1302**. However, it should be noted that in alternative embodiments, the condition being regulated by the control system **1300** may be correlative to the measured values obtained by the sensing arrangement **1304**. That said, for clarity and purposes of explanation, the subject matter may be described herein in the context of the sensing arrangement **1304** being realized as a glucose sensing arrangement that senses, detects, measures or otherwise quantifies the user's glucose level, which is being regulated in the body **1301** of the user by the control system **1300**.

[0116] In exemplary embodiments, the sensing arrangement **1304** includes one or more interstitial glucose sensing elements that generate or otherwise output electrical signals having a signal characteristic that is correlative to, influenced by, or otherwise indicative of the relative interstitial fluid glucose level in the body **1301** of the user. The output electrical signals are filtered or otherwise processed to obtain a measurement value indicative of the user's interstitial fluid glucose level. In exemplary embodiments, a blood glucose meter **1330**, such as a finger stick device, is utilized to directly sense, detect, measure or otherwise quantify the blood glucose in the body **1301** of the user. In this regard, the blood glucose meter **1330** outputs or otherwise provides a measured blood glucose value that may be utilized as a reference measurement for calibrating the sensing arrangement **1304** and converting a measurement value indicative of the user's interstitial fluid glucose level into a corresponding calibrated blood glucose value. For purposes of explanation, the calibrated blood glucose value calculated based on the electrical signals output by the sensing element(s) of the sensing arrangement **1304** may alternatively be referred to herein as the sensor glucose value, the sensed glucose value, or variants thereof.

[0117] In the illustrated embodiment, the pump control system **1320** generally represents the electronics and other components of the infusion device **1302** that control operation of the fluid infusion device **1302** according to a desired infusion delivery program in a manner that is influenced by the sensed glucose value indicative of a current glucose level in the body **1301** of the user. For example, to support a closed-loop operating mode, the pump control system **1320** maintains, receives, or otherwise obtains a target or commanded glucose value, and automatically generates or otherwise determines dosage commands for operating an actuation arrangement, such as a motor **1307**, to displace the plunger **1317** and deliver insulin to the body **1301** of the user based on the difference between a sensed glucose value and the target glucose value. In other operating modes, the pump control system **1320** may generate or otherwise determine dosage commands configured to maintain the sensed glucose value below an upper glucose limit, above a lower glucose limit, or otherwise within a desired range of glucose values. In practice, the infusion device **1302** may store or otherwise maintain the target value, upper and/or lower glucose limit(s), and/or other glucose threshold value(s) in a data storage element accessible to the pump control

system **1320**.

[0118] The target glucose value and other threshold glucose values may be received from an external component (e.g., CCD **906** and/or computing device **908**) or be input by a user via a user interface element **1340** associated with the infusion device **1302**. In practice, the one or more user interface element(s) **1340** associated with the infusion device **1302** typically include at least one input user interface element, such as, for example, a button, a keypad, a keyboard, a knob, a joystick, a mouse, a touch panel, a touchscreen, a microphone or another audio input device, and/or the like. Additionally, the one or more user interface element(s) **1340** include at least one output user interface element, such as, for example, a display element (e.g., a light-emitting diode or the like), a display device (e.g., a liquid crystal display or the like), a speaker or another audio output device, a haptic feedback device, or the like, for providing notifications or other information to the user. It should be noted that although FIG. **13** depicts the user interface element(s) **1340** as being separate from the infusion device **1302**, in practice, one or more of the user interface element(s) **1340** may be integrated with the infusion device **1302**. Furthermore, in some embodiments, one or more user interface element(s) **1340** are integrated with the sensing arrangement **1304** in addition to and/or in alternative to the user interface element(s) **1340** integrated with the infusion device **1302**. The user interface element(s) **1340** may be manipulated by the user to operate the infusion device **1302** to deliver correction boluses, adjust target and/or threshold values, modify the delivery control scheme or operating mode, and the like, as desired.

[0119] Still referring to FIG. **13**, in the illustrated embodiment, the infusion device **1302** includes a motor control module **1312** coupled to a motor **1307** (e.g., motor assembly **1007**) that is operable to displace a plunger **1317** (e.g., plunger **1017**) in a reservoir (e.g., reservoir **1005**) and provide a desired amount of fluid to the body **1301** of a user. In this regard, displacement of the plunger **1317** results in the delivery of a fluid that is capable of influencing the condition in the body **1301** of the user to the body **1301** of the user via a fluid delivery path (e.g., via tubing **1021** of an infusion set **1025**). A motor driver module **1314** is coupled between an energy source **1303** and the motor **1307**. The motor control module **1312** is coupled to the motor driver module **1314**, and the motor control module **1312** generates or otherwise provides command signals that operate the motor driver module **1314** to provide current (or power) from the energy source **1303** to the motor **1307** to displace the plunger **1317** in response to receiving, from a pump control system **1320**, a dosage command indicative of the desired amount of fluid to be delivered.

[0120] In exemplary embodiments, the energy source **1303** is realized as a battery housed within the infusion device **1302** (e.g., within housing **1002**) that provides direct current (DC) power. In this regard, the motor driver module **1314** generally represents the combination of circuitry, hardware and/or other electrical components configured to convert or otherwise transfer DC power provided by the energy source **1303** into alternating electrical signals applied to respective phases of the stator windings of the motor **1307** that result in current flowing through the stator windings that generates a stator magnetic field and causes the rotor of the motor **1307** to rotate. The motor control module **1312** is configured to receive or otherwise obtain a commanded dosage from the pump control system **1320**, convert the commanded dosage to a commanded translational displacement of the plunger **1317**, and command, signal, or otherwise operate the motor driver module **1314** to cause the rotor of the motor **1307** to rotate by an amount that produces the commanded translational displacement of the plunger **1317**. For example, the motor control module **1312** may determine an amount of rotation of the rotor required to produce translational displacement of the plunger **1317** that achieves the commanded dosage received from the pump control system **1320**. Based on the current rotational position (or orientation) of the rotor with respect to the stator that is indicated by the output of the rotor sensing arrangement **1316**, the motor control module **1312** determines the appropriate sequence of alternating electrical signals to be applied to the respective phases of the stator windings that should rotate the rotor by the determined amount of rotation from its current position (or orientation). In embodiments where the motor **1307**

is realized as a BLDC motor, the alternating electrical signals commutate the respective phases of the stator windings at the appropriate orientation of the rotor magnetic poles with respect to the stator and in the appropriate order to provide a rotating stator magnetic field that rotates the rotor in the desired direction. Thereafter, the motor control module **1312** operates the motor driver module **1314** to apply the determined alternating electrical signals (e.g., the command signals) to the stator windings of the motor **1307** to achieve the desired delivery of fluid to the user.

[0121] When the motor control module **1312** is operating the motor driver module **1314**, current flows from the energy source **1303** through the stator windings of the motor **1307** to produce a stator magnetic field that interacts with the rotor magnetic field. In some embodiments, after the motor control module **1312** operates the motor driver module **1314** and/or motor **1307** to achieve the commanded dosage, the motor control module **1312** ceases operating the motor driver module **1314** and/or motor **1307** until a subsequent dosage command is received. In this regard, the motor driver module **1314** and the motor **1307** enter an idle state during which the motor driver module **1314** effectively disconnects or isolates the stator windings of the motor **1307** from the energy source **1303**. In other words, current does not flow from the energy source **1303** through the stator windings of the motor **1307** when the motor **1307** is idle, and thus, the motor **1307** does not consume power from the energy source **1303** in the idle state, thereby improving efficiency.

[0122] Depending on the embodiment, the motor control module **1312** may be implemented or realized with a general purpose processor, a microprocessor, a controller, a microcontroller, a state machine, a content addressable memory, an application specific integrated circuit, a field programmable gate array, any suitable programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof, designed to perform the functions described herein. In exemplary embodiments, the motor control module **1312** includes or otherwise accesses a data storage element or memory, including any sort of random access memory (RAM), read only memory (ROM), flash memory, registers, hard disks, removable disks, magnetic or optical mass storage, or any other short or long term storage media or other non-transitory computer-readable medium, which is capable of storing programming instructions for execution by the motor control module **1312**. The computer-executable programming instructions, when read and executed by the motor control module **1312**, cause the motor control module **1312** to perform or otherwise support the tasks, operations, functions, and processes described herein.

[0123] It should be appreciated that FIG. **13** is a simplified representation of the infusion device **1302** for purposes of explanation and is not intended to limit the subject matter described herein in any way. In this regard, depending on the embodiment, some features and/or functionality of the sensing arrangement **1304** may implemented by or otherwise integrated into the pump control system **1320**, or vice versa. Similarly, in practice, the features and/or functionality of the motor control module **1312** may implemented by or otherwise integrated into the pump control system **1320**, or vice versa. Furthermore, the features and/or functionality of the pump control system **1320** may be implemented by control electronics **1024** located in the fluid infusion device **1302**, while in alternative embodiments, the pump control system **1320** may be implemented by a remote computing device that is physically distinct and/or separate from the infusion device **1302**, such as, for example, the CCD **906** or the computing device **908**.

[0124] FIG. **14** depicts an exemplary embodiment of a pump control system **1400** suitable for use as the pump control system **1320** in FIG. **13** in accordance with one or more embodiments. The illustrated pump control system **1400** includes, without limitation, a pump control module **1402**, a communications interface **1404**, and a data storage element (or memory) **1406**. The pump control module **1402** is coupled to the communications interface **1404** and the memory **1406**, and the pump control module **1402** is suitably configured to support the operations, tasks, and/or processes described herein. In exemplary embodiments, the pump control module **1402** is also coupled to one or more user interface elements **1408** (e.g., user interface **1030**, **1340**) for receiving user input and providing notifications, alerts, or other therapy information to the user. Although FIG. **14** depicts

the user interface element **1408** as being separate from the pump control system **1400**, in various alternative embodiments, the user interface element **1408** may be integrated with the pump control system **1400** (e.g., as part of the infusion device **1302**), the sensing arrangement **1304** or another element of an infusion system **900** (e.g., the computer **908** or CCD **906**).

[0125] Referring to FIG. **14** and with reference to FIG. **13**, the communications interface **1404** generally represents the hardware, circuitry, logic, firmware and/or other components of the pump control system **1400** that are coupled to the pump control module **1402** and configured to support communications between the pump control system **1400** and the sensing arrangement **1304**. In this regard, the communications interface **1404** may include or otherwise be coupled to one or more transceiver modules capable of supporting wireless communications between the pump control system **1320**, **1400** and the sensing arrangement **1304** or another electronic device **106**, **110**, **800**, **812**, **906**, **908** in an infusion system **900** or a management system **100**, **816**. For example, the communications interface **1404** may be utilized to receive sensor measurement values or other measurement data from a sensing arrangement **904**, **1304** as well as upload such sensor measurement values to a server **106** or other computing device **110**, **800**, **812**, **908** for purposes of detecting meals or other events and generating reports and related GUI displays as described above. In other embodiments, the communications interface **1404** may be configured to support wired communications to/from the sensing arrangement **1304**.

[0126] The pump control module **1402** generally represents the hardware, circuitry, logic, firmware and/or other component of the pump control system **1400** that is coupled to the communications interface **1404** and configured to determine dosage commands for operating the motor **1306** to deliver fluid to the body **1301** based on data received from the sensing arrangement **1304** and perform various additional tasks, operations, functions and/or operations described herein. For example, in exemplary embodiments, pump control module **1402** implements or otherwise executes a command generation application **1410** that supports one or more autonomous operating modes and calculates or otherwise determines dosage commands for operating the motor **1306** of the infusion device **1302** in an autonomous operating mode based at least in part on a current measurement value for a condition in the body **1301** of the user. For example, in a closed-loop operating mode, the command generation application **1410** may determine a dosage command for operating the motor **1306** to deliver insulin to the body **1301** of the user based at least in part on the current glucose measurement value most recently received from the sensing arrangement **1304** to regulate the user's blood glucose level to a target reference glucose value. Additionally, the command generation application **1410** may generate dosage commands for boluses that are manually-initiated or otherwise instructed by a user via a user interface element **1408**.

[0127] Still referring to FIG. **14**, depending on the embodiment, the pump control module **1402** may be implemented or realized with a general purpose processor, a microprocessor, a controller, a microcontroller, a state machine, a content addressable memory, an application specific integrated circuit, a field programmable gate array, any suitable programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof, designed to perform the functions described herein. In this regard, the steps of a method or algorithm described in connection with the embodiments disclosed herein may be embodied directly in hardware, in firmware, in a software module executed by the pump control module **1402**, or in any practical combination thereof. In exemplary embodiments, the pump control module **1402** includes or otherwise accesses the data storage element or memory **1406**, which may be realized using any sort of non-transitory computer-readable medium capable of storing programming instructions for execution by the pump control module **1402**. The computer-executable programming instructions, when read and executed by the pump control module **1402**, cause the pump control module **1402** to implement or otherwise generate the command generation application **1410** and perform tasks, operations, functions, and processes described herein.

[0128] It should be understood that FIG. **14** is a simplified representation of a pump control system

1400 for purposes of explanation and is not intended to limit the subject matter described herein in any way. For example, in some embodiments, the features and/or functionality of the motor control module **1312** may be implemented by or otherwise integrated into the pump control system **1400** and/or the pump control module **1402**, for example, by the command generation application **1410** converting the dosage command into a corresponding motor command, in which case, the separate motor control module **1312** may be absent from an embodiment of the infusion device **1302**.

[0129] FIG. **15** depicts an exemplary closed-loop control system **1500** that may be implemented by a pump control system **1320**, **1400** to provide a closed-loop operating mode that autonomously regulates a condition in the body of a user to a reference (or target) value. It should be appreciated that FIG. **15** is a simplified representation of the control system **1500** for purposes of explanation and is not intended to limit the subject matter described herein in any way.

[0130] In exemplary embodiments, the control system **1500** receives or otherwise obtains a target glucose value at input **1502**. In some embodiments, the target glucose value may be stored or otherwise maintained by the infusion device **1302** (e.g., in memory **1406**), however, in some alternative embodiments, the target value may be received from an external component (e.g., CCD **906** and/or computer **908**). In one or more embodiments, the target glucose value may be dynamically calculated or otherwise determined prior to entering the closed-loop operating mode based on one or more patient-specific control parameters. For example, the target blood glucose value may be calculated based at least in part on a patient-specific reference basal rate and a patient-specific daily insulin requirement, which are determined based on historical delivery information over a preceding interval of time (e.g., the amount of insulin delivered over the preceding 24 hours). The control system **1500** also receives or otherwise obtains a current glucose measurement value (e.g., the most recently obtained sensor glucose value) from the sensing arrangement **1304** at input **1504**. The illustrated control system **1500** implements or otherwise provides proportional-integral-derivative (PID) control to determine or otherwise generate delivery commands for operating the motor **1310** based at least in part on the difference between the target glucose value and the current glucose measurement value. In this regard, the PID control attempts to minimize the difference between the measured value and the target value, and thereby regulates the measured value to the desired value. PID control parameters are applied to the difference between the target glucose level at input **1502** and the measured glucose level at input **1504** to generate or otherwise determine a dosage (or delivery) command provided at output **1530**. Based on that delivery command, the motor control module **1312** operates the motor **1310** to deliver insulin to the body of the user to influence the user's glucose level, and thereby reduce the difference between a subsequently measured glucose level and the target glucose level.

[0131] The illustrated control system **1500** includes or otherwise implements a summation block **1506** configured to determine a difference between the target value obtained at input **1502** and the measured value obtained from the sensing arrangement **1304** at input **1504**, for example, by subtracting the target value from the measured value. The output of the summation block **1506** represents the difference between the measured and target values, which is then provided to each of a proportional term path, an integral term path, and a derivative term path. The proportional term path includes a gain block **1520** that multiplies the difference by a proportional gain coefficient, $K_{sub.P}$, to obtain the proportional term. The integral term path includes an integration block **1508** that integrates the difference and a gain block **1522** that multiplies the integrated difference by an integral gain coefficient, $K_{sub.I}$, to obtain the integral term. The derivative term path includes a derivative block **1510** that determines the derivative of the difference and a gain block **1524** that multiplies the derivative of the difference by a derivative gain coefficient, $K_{sub.D}$, to obtain the derivative term. The proportional term, the integral term, and the derivative term are then added or otherwise combined to obtain a delivery command that is utilized to operate the motor at output **1530**. Various implementation details pertaining to closed-loop PID control and determine gain coefficients are described in greater detail in U.S. Pat. No. 7,402,153, which is incorporated by

reference.

[0132] In one or more exemplary embodiments, the PID gain coefficients are user-specific (or patient-specific) and dynamically calculated or otherwise determined prior to entering the closed-loop operating mode based on historical insulin delivery information (e.g., amounts and/or timings of previous dosages, historical correction bolus information, or the like), historical sensor measurement values, historical reference blood glucose measurement values, user-reported or user-input events (e.g., meals, exercise, and the like), and the like. In this regard, one or more patient-specific control parameters (e.g., an insulin sensitivity factor, a daily insulin requirement, an insulin limit, a reference basal rate, a reference fasting glucose, an active insulin action duration, pharmacodynamical time constants, or the like) may be utilized to compensate, correct, or otherwise adjust the PID gain coefficients to account for various operating conditions experienced and/or exhibited by the infusion device **1302**. The PID gain coefficients may be maintained by the memory **1406** accessible to the pump control module **1402**. In this regard, the memory **1406** may include a plurality of registers associated with the control parameters for the PID control. For example, a first parameter register may store the target glucose value and be accessed by or otherwise coupled to the summation block **1506** at input **1502**, and similarly, a second parameter register accessed by the proportional gain block **1520** may store the proportional gain coefficient, a third parameter register accessed by the integration gain block **1522** may store the integration gain coefficient, and a fourth parameter register accessed by the derivative gain block **1524** may store the derivative gain coefficient.

[0133] Referring to FIGS. **13-15**, in one or more embodiments, the patient-specific meal detection model may be stored or otherwise maintained by the pump control system **1320**, **1400** (e.g., in memory **1406**) to support the infusion device **1302** performing the meal detection process **600** substantially in real-time. In this regard, the pump control system **1320**, **1400** may continually monitor the sensor glucose measurement values received from the sensing arrangement **1304** and apply the meal detection model to the most recent set of sensor glucose measurement values (e.g., buffered sensor glucose measurements corresponding to the preceding hour) to detect or otherwise identify whether the sensor glucose measurements indicate that the patient has consumed a meal. In response to detecting a meal, the pump control system **1320**, **1400** may tag, mark, or otherwise provide an indication associated with one or more sensor glucose measurement values prior to uploading the sensor glucose measurement values to a server (e.g., server **106**). Additionally, in some embodiments, the pump control system **1320**, **1400** may automatically adjust one or more aspects of the autonomous operation of the infusion device **1302** to temporarily modify delivery of the infusion device **1302** to account for the patient consuming a meal. For example, one or more of the closed-loop gain coefficients **1520**, **1522**, **1524** may be temporarily modified to alter the response of the closed-loop control system **1500** in a manner that more effectively regulates or moderates the patient's glucose level after consuming a meal (e.g., to prevent a postprandial hyperglycemic or hypoglycemic event). In this regard, it should be appreciated that the patient-specific meal detection models may be employed in any number of different ways depending on the needs of a particular application, and the subject matter described herein is not limited to any particular implementation described herein.

[0134] It should be noted that although the subject matter may be described herein primarily in the context of an infusion device delivering insulin to the body of a patient with diabetes to regulate the patient's glucose level for purposes of explanation, in practice, the subject matter is not limited to use with infusion devices, insulin, diabetes or glucose control, and the like. Rather, the subject matter may be implemented in an equivalent manner in the context of patient management systems that do not include an infusion device, for example, in systems with where patients self-administer injections, oral medications, or the like, in systems where a sensing arrangement is utilized to monitor any a physiological condition of a patient in a substantially continuous manner, or in the context of a patient with dysglycemia or another physiological condition being monitored that is

influenced by meals or other behavioral events.

[0135] For the sake of brevity, conventional techniques related to glucose sensing and/or monitoring, bolusing, meal boluses or correction boluses, and other functional aspects of the subject matter may not be described in detail herein. In addition, certain terminology may also be used in the herein for the purpose of reference only, and thus is not intended to be limiting. For example, terms such as “first”, “second”, and other such numerical terms referring to structures do not imply a sequence or order unless clearly indicated by the context. The foregoing description may also refer to elements or nodes or features being “connected” or “coupled” together. As used herein, unless expressly stated otherwise, “coupled” means that one element/node/feature is directly or indirectly joined to (or directly or indirectly communicates with) another element/node/feature, and not necessarily mechanically.

[0136] While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or embodiments described herein are not intended to limit the scope, applicability, or configuration of the claimed subject matter in any way. For example, the subject matter described herein is not necessarily limited to the infusion devices and related systems described herein. Moreover, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the described embodiment or embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope defined by the claims, which includes known equivalents and foreseeable equivalents at the time of filing this patent application. Accordingly, details of the exemplary embodiments or other limitations described above should not be read into the claims absent a clear intention to the contrary.

Claims

1. A method comprising: obtaining a plurality of glucose measurements indicative of a glucose level in a body of a patient during an analysis interval; detecting an occurrence of a meal during the analysis interval based on the plurality of glucose measurements obtained during the analysis interval; and updating a user interface to include an indication of the meal detected during the analysis interval.
2. The method of claim 1, wherein the user interface includes a plurality of indications each corresponding to a meal, and wherein at least one indication of the plurality of indications corresponds to an announced meal.
3. The method of claim 1, wherein the indication of the meal is presented overlaid on a representation of the plurality of glucose measurements.
4. The method of claim 3, further comprising presenting an indication of a target glucose range in conjunction with the representation of the plurality of glucose measurements.
5. The method of claim 3, wherein the user interface comprises a graph that presents glucose measurements, including the plurality of glucose measurements, as a function of time.
6. The method of claim 1, further comprising configuring a monitoring period for a particular type of meal to detect future occurrences of the particular type of meal based on the detected occurrence of the meal.
7. The method of claim 6, further comprising presenting an indication of the monitoring period within the user interface.
8. The method of claim 1, further comprising: detecting a pattern based on the plurality of glucose measurements; and presenting an indication of the detected pattern within the user interface.
9. The method of claim 1, wherein detecting the occurrence of the meal during the analysis interval comprises determining one or more glucose measurement statistics based on the plurality of glucose measurements, wherein the occurrence of the meal is detected by determining that the one

or more glucose measurement statistics are correlative to an occurrence of a meal for the patient.

10. A system comprising: one or more processors; and one or more processor-readable storage media storing instructions which, when executed by the one or more processors, cause performance of: obtaining a plurality of glucose measurements indicative of a glucose level in a body of a patient during an analysis interval; detecting an occurrence of a meal during the analysis interval based on the plurality of glucose measurements obtained during the analysis interval; and updating a user interface to include an indication of the meal detected during the analysis interval.

11. The system of claim 10, wherein the user interface includes a plurality of indications each corresponding to a meal, and wherein at least one indication of the plurality of indications corresponds to an announced meal.

12. The system of claim 10, wherein the indication of the meal is presented overlaid on a representation of the plurality of glucose measurements.

13. The system of claim 12, wherein the instructions further cause performance of presenting an indication of a target glucose range in conjunction with the representation of the plurality of glucose measurements.

14. The system of claim 12, wherein the user interface comprises a graph that presents glucose measurements, including the plurality of glucose measurements, as a function of time.

15. The system of claim 10, wherein the instructions further cause performance of configuring a monitoring period for a particular type of meal to detect future occurrences of the particular type of meal based on the detected occurrence of the meal.

16. The system of claim 15, wherein the instructions further cause performance of presenting an indication of the monitoring period within the user interface.

17. The system of claim 10, wherein the instructions further cause performance of: detecting a pattern based on the plurality of glucose measurements; and presenting an indication of the detected pattern within the user interface.

18. The system of claim 10, wherein detecting the occurrence of the meal during the analysis interval comprises determining one or more glucose measurement statistics based on the plurality of glucose measurements, wherein the occurrence of the meal is detected by determining that the one or more glucose measurement statistics are correlative to an occurrence of a meal for the patient.

19. A method comprising: obtaining a plurality of glucose measurements indicative of a glucose level in a body of a patient during an analysis interval; detecting an occurrence of a meal during the analysis interval based on the plurality of glucose measurements obtained during the analysis interval; and presenting a meal marker icon in a user interface that includes a graph presenting glucose measurements including the plurality of glucose measurements, wherein the meal marker icon is overlaid on the graph.

20. The method of claim 1, wherein the user interface further comprises a second meal marker icon overlaid on the graph, the second meal marker icon corresponding to an announced meal.
